

Notes

Chapter 1. Introduction

1 To a considerable extent this is also true of physical science; see P. Janich, "Physics – natural science or technology?" in W. Krohn, E. T. Layton, Jr., and P. Weingart, eds., *The Dynamics of Science and Technology* (Dordrecht: Reidel, 1978), pp. 3–27.

2 For a general critique of the distinction between the two contexts, see P. Feyrabend, *Against Method* (London: NLB, 1975), pp. 165–169; and T. Nickles, "Scientific discovery and the future of philosophy of science," in T. Nickles, ed., *Scientific Discovery, Logic and Rationality*, Boston Studies in the Philosophy of Science, vol. 56 (Dordrecht and Boston: Reidel, 1980), pp. 1–59.

3 See S. Bem, "Context of discovery and conceptual history of psychology," in S. Bem, H. Rappard, and W. van Hoorn, eds., *Proceedings of the First European Meeting of Cheiron* (Leiden: Ryksuniversiteit, 1983), pp. 207–231.

4 "Discoveries do not simply 'occur' or 'happen' naturalistically, but are socially defined and recognized productions. In other words, the question is not what makes them happen, but rather what makes them discoveries." See A. Brannigan, *The Social Basis of Scientific Discoveries* (Cambridge: Cambridge University Press, 1981), p. 77.

5 cf. K. J. Gergen, *Toward Transformation in Social Knowledge* (New York: Springer, 1982), p. 101: "The audience for research reports is never exposed to ongoing events; one never gains first-hand experience with the research process itself. Rather, the chief product of research is language. Research reports essentially furnish linguistic accounts or interpretations."

6 A useful analysis of such constructive schemes, with illustrations from the history of experimental psychology, has been presented by Gernot Böhme, "The social function of cognitive structures: A concept of the scientific community within a theory of action," in K. D. Knorr, H. Strasser, and H. G. Zillian, eds., *Determinants and Controls of Scientific Development* (Dordrecht: Reidel, 1975), pp. 205–225; and G. Böhme, "Cognitive norms, knowledge interests and the constitution of the scientific object: A case study in the functioning of rules for experimentation," in E. Mendelsohn, P. Weingart, and R. Whitley, eds., *The Social Production of Scientific Knowledge* (Dordrecht: Reidel, 1977), pp. 129–141.

7 Psychological inquiry now acquired the general features of scientific inquiry insofar as the latter can be characterized as “a special sort of craft work which operates on intellectually constructed objects.” See T. R. Ravetz, *Scientific Knowledge and Its Social Problems* (New York: Oxford University Press, 1971), p. 116.

8 It can be argued that the same situation exists in respect to any system of purely logical procedures: “As a body of conventions and esoteric traditions the compelling character of logic, such as it is, derives from certain narrowly defined purposes and from custom and institutionalized usage.” See B. Barnes and D. Bloor, “Relativism, rationalism and the sociology of knowledge,” in M. Hollis and S. Lukes, eds., *Rationality and Relativism* (Oxford: Blackwell, 1982), pp. 21–47 (p. 45).

9 G. Devereux, *From Anxiety to Method in the Behavioral Sciences* (The Hague: Mouton, 1967). Devereux was the first to recognize the depth of anxiety surrounding the area of methodology as an indication of fundamental problems in the behavioral sciences. However, he saw the sources of this anxiety only in terms of problems arising for the individual investigator in his relationship to the human subject of his investigation. The problems of investigative practice therefore appeared analogous to the problems of psychoanalytic practice. This radically individualist framework produced a complete blindness toward the collective anxieties of groups of investigators arising out of real or imagined threats to their position in the scientific status structure.

10 Some important consequences are discussed in R. D. Romanynshyn, “Method and meaning in psychology: The method has been the message,” *Journal of Phenomenological Psychology* 2 (1971):93–113; and C. Argyris, *Inner Contradictions of Rigorous Research* (New York: Academic Press, 1980).

11 The scope of the profound changes that have occurred in this field is perhaps most readily appreciated by the outsider through a comparison of a text representing the old approach, like J. Ben-David, *The Scientist's Role in Society* (Englewood Cliffs, N.J.: Prentice-Hall, 1971), and one informed by more recent developments, like J. Law and P. Lodge, *Science for Social Scientists* (London: Macmillan, 1984). For a succinct introduction to issues in this field, see the introduction to K. D. Knorr-Cetina and M. Mulkay, eds., *Science Observed: Perspectives on the Social Study of Science* (London and Beverly Hills, Calif.: Sage, 1983), pp. 1–17.

12 E.g., A. R. Buss, ed., *Psychology in Social Context* (New York: Irvington, 1979).

13 The existence of a long-unheeded exception drives home its complete isolation in the literature; see S. Rosenzweig, “The experimental situation as a psychological problem,” *Psychological Review* 40 (1933):337–354. Because of their exposure to real world problems, industrial psychologists were forced to recognize social features in investigative situations at an early stage; see W. Schulte, “Untersuchungen über den Einfluss des Versuchsleiters auf das Prüfergebnis,” *Industrielle Psychotechnik* 1 (1924):289–291; and Elton Mayo, *The Human Problems of an Industrial Civilization* (New York: Macmillan, 1933). However, the extension of these insights from the “profane” area of industrial settings to the “sacred” precincts of university laboratories seems to have encountered formidable cognitive and affective barriers. A parallel case occurred in the field of animal experimentation; see D. Fernald, *The Hans Legacy: A Story of Science* (Hillsdale, N.J., and London: Erlbaum, 1984).

14 Discussions of this literature are found in T. X. Barber, *Pitfalls in Human Research: Ten Pivotal Points* (New York: Pergamon Press, 1976); R. Rosenthal, *Experimenter Effects in Behavioral Research*, rev. ed. (New York: Irvington, 1976); R. Rosenthal and R. L. Rosnow, eds., *Artifact in Behavioral Research* (New York:

Academic Press, 1969); R. Rosenthal and R. L. Rosnow, *The Volunteer Subject* (New York: Wiley, 1975); R. Rosenthal and D. B. Rubin, "Interpersonal expectancy effects: The first 345 studies," *The Behavioral and Brain Sciences* 3 (1978):377–415; J. M. Suls and R. L. Rosnow, "Concerns about artifacts in psychological experiments," in J. G. Morawski, ed., *The Rise of Experimentation in American Psychology* (New Haven: Yale University Press, 1988), pp. 163–187.

15 J. G. Adair, "Commentary," *The Behavioral and Brain Sciences* 3 (1978):386.

16 R. M. Farr, "On the social significance of artifacts in experimenting," *British Journal of Social and Clinical Psychology* 17 (1978):299–306.

17 L. W. Brandt, "Experimenter-effect research," *Psychologische Beiträge* 17 (1975):133–140.

18 See, however, J. G. Adair, *The Human Subject: The Social Psychology of the Psychological Experiment* (Boston: Little, Brown, 1973); and N. Friedman, *The Social Nature of Psychological Research: The Psychological Experiment as a Social Interaction* (New York: Basic Books, 1967).

19 D. P. Schultz, "The nature of the human data source in psychology," in D. P. Schultz, ed., *The Science of Psychology: Critical Reflections* (New York: Appleton-Century-Crofts, 1970), pp. 77–86.

20 K. Holzkamp, *Kritische Psychologie* (Frankfurt: Fischer, 1972); W. Mertens, *Sozialpsychologie des Experiments* (Hamburg: Hoffman & Campe, 1975); W. Bungard, ed., *Die gute Versuchsperson denkt nicht: Artefakte in der Sozialpsychologie* (Munich and Baltimore: Urban & Schwarzenberg, 1980).

21 H. M. Collins, *Changing Order: Replication and Induction in Scientific Practice* (London and Beverly Hills, Calif.: Sage, 1985). See also K. Knorr, R. Krohn, and R. Whitley, eds., *The Social Process of Scientific Investigation, Sociology of the Sciences Yearbook*, vol. 4 (Boston and Dordrecht: Reidel, 1980); and K. D. Knorr-Cetina, *The Manufacture of Knowledge: An Essay on the Constructivist and Contextual Nature of Science* (Oxford: Pergamon Press, 1981).

22 Moreover, there are inherent limitations on the use of experimental methods to investigate the experimental method; see H. Gadlin and G. Ingle, "Through the one-way mirror: The limits of experimental self-reflection," *American Psychologist* 30 (1975):1003–1009.

23 R. Whitley, *The Intellectual and Social Organization of the Sciences* (Oxford: Clarendon Press, 1984).

24 T. M. Ziman, *Public Knowledge: An Essay concerning the Social Dimension of Science* (Cambridge: Cambridge University Press, 1968).

25 E. B. Titchener, "Psychology: Science or technology?" *Popular Science Monthly* 84 (1914):39–51.

26 Because of their pedagogical mission, textbook histories frequently suggest some such position; see M. G. Ash, "The self-presentation of a discipline: History of psychology in the United States between pedagogy and scholarship," in L. Graham, W. Lepenies, and P. Weingart, eds., *Functions and Uses of Disciplinary Histories, Sociology of the Sciences Yearbook*, vol. 7 (Dordrecht: Reidel, 1983), pp. 143–189.

27 On the justificationist implications of this position, see W. B. Weimer, "The history of psychology and its retrieval from historiography: I. The problematic nature of history," *Science Studies* 4 (1974):235–258.

28 This recognition depended on an emphasis on the craft character of science by M. Polyani, *Personal Knowledge* (London: Routledge and Kegan Paul, 1958). The

implications of such an analysis were extensively developed by Ravetz, *Scientific Knowledge*.

29 These kinds of studies have proliferated greatly in recent years. A well-known early example of the observational study of scientific work was B. Latour and S. Woolgar, *Laboratory Life: The Social Construction of Scientific Facts* (London and Beverly Hills, Calif.: Sage, 1979). The use of interview material is exemplified by G. N. Gilbert and M. Mulkay, *Opening Pandora's Box: A Sociological Analysis of Scientists' Discourse* (Cambridge: Cambridge University Press, 1984).

30 For some sociologists of science the existence of this distinction appears to have provided the occasion for claiming a privileged status for either the direct observation of practice or the analysis of discourse about practice. The latter position has been forcefully argued in G. N. Gilbert and M. Mulkay, "Experiments are the key: Participants' histories and historians' histories of science," *Isis* 75 (1984):105–125 (but see S. Shapin, "Talking history: Reflections on discourse analysis," *Isis* 75 [1984]:125–130). However, the production of a certain kind of discourse in the form of scientific publications lies at the core of scientific practice itself. If we wish to follow the historical development of science, we should be looking at this discourse. But because this discourse is part of the ordinary practice of science, the useful distinction to make here is that between unreflected practice and reflections on that practice, not between "discourse" on the one hand and actions and beliefs on the other.

31 A good example of the use of scientific articles as historical documents is provided by C. Bazerman, "Modern evolution of the experimental report in physics: Spectroscopic articles in *Physical Review*, 1893–1980," *Social Studies of Science* 14 (1984):163–196. See also C. Bazerman, *Shaping Written Knowledge: The Genre and Activity of the Written Article in Science* (Madison: University of Wisconsin Press, 1988).

32 Two major exceptions are J. S. Bruner and G. W. Allport, "Fifty years of change in American psychology," *Psychological Bulletin* 37 (1940):757–776; and R. W. Lissitz, "A longitudinal study of the research methodology in the Journal of Abnormal and Social Psychology, the Journal of Nervous and Mental Disease and the American Journal of Psychiatry," *Journal of the History of the Behavioral Sciences* 5 (1969):248–255.

33 A comprehensive review of the use of the second type of analysis is provided by C. Bazerman, "Scientific writing as a social act: Review of the literature of the sociology of science," in P. V. Anderson, R. J. Brockmann, and C. R. Miller, eds., *New Essays in Technical and Scientific Communication: Research, Theory, Practice* (Farmingdale, N.Y.: Baywood, 1983), pp. 156–184. Once the informal rules governing the presentation of the knowledge product have become formalized they can also be directly analyzed. See C. Bazerman, "Codifying the social scientific style: The APA Publication Manual as a Behaviorist Rhetoric," in J. S. Nelson, A. Megill, D. N. McCloskey, eds., *The Rhetoric of the Human Sciences* (Madison: University of Wisconsin Press, 1987), pp. 125–144.

34 On the relationship between research publications and research practices, see F. L. Holmes, "Scientific writing and scientific discovery," *Isis* 78 (1987):220–235.

35 See S. Toulmin and D. E. Leary, "The cult of empiricism in psychology and beyond," in S. Koch and D. E. Leary, eds., *A Century of Psychology as Science: Retrospectives and Assessments* (New York: McGraw-Hill, 1985), pp. 594–617.

36 This issue is explored further in K. Danziger, "Towards a conceptual framework for a critical history of psychology," in H. Carpintero and J. M. Peiro, eds.,

Psychology in Its Historical Context: Essays in Honour of Prof. Josef Brozek (Valencia: Monografías de la Revista de Historia de la Psicología, 1984), pp. 99–107.

Chapter 2. Historical roots of the psychological laboratory

1 See W. G. Bringmann, N. J. Bringmann, and G. A. Ungerer, “The establishment of Wundt’s laboratory: An archival and documentary study,” in W. G. Bringmann and R. D. Tweney, eds., *Wundt Studies: A Centennial Collection* (Toronto: C. J. Hogrefe, 1980), pp. 123–157.

2 Empiricist mental philosophy did of course proceed in a methodical way from complex ideas to “simple” ones that had to be isolated in experience, and this according to Locke “requires pains and assiduity” (*An Essay concerning Human Understanding*, bk. 2, chap. 13, sec. 28), but this was a general prescription, not related to the special problems of the “inner sense.”

3 The closest that even an archintrospectionist like James Mill comes to methodological self-reflection is when he advises that “the learner should by practice acquire the habit of reflecting upon his Sensations, as a distinct class of feelings.” See J. Mill, *Analysis of the Phenomena of the Human Mind* (1829; repr. New York: A. M. Kelley, 1967), p. 2.

4 The most important examples are provided by John Stuart Mill’s defense of introspection in his *Auguste Comte and Positivism* (London: G. Routledge, [1865]), and William Hamilton’s *Lectures on Metaphysics and Logic* (Edinburgh: Blackwood, 1859), lecture 19, which is strongly influenced by his study of Kant.

5 Immanuel Kant, *Die metaphysischen Anfangsgründe der Naturwissenschaft* (1786); see esp. p. 8 of the English translation, *Metaphysical Foundations of Natural Science*, trans. J. Ellington (Indianapolis: Bobbs-Merrill, 1970). To get the measure of Kant’s new methodological sensitivity one need only compare his treatment of the evidential value of the inner sense with that of his guide in matters psychological, Johann Nicolas Tetens. See J. N. Tetens, *Philosophische Versuche über die menschliche Natur und ihre Entwicklung* (Leipzig: Weidmann u. Reich, 1777). Tetens in turn shows more awareness of methodological problems than his predecessors.

6 For a good modern discussion of this issue, see R. Rorty, *Philosophy and the Mirror of Nature* (Princeton: Princeton University Press, 1979), chap. 3, sec. 2.

7 On Kant and psychology, see T. Mischel, “Kant and the possibility of a science of psychology,” *Monist* 51 (1967):599–622; C. Gouaux, “Kant’s view on the nature of empirical psychology,” *Journal of the History of the Behavioral Sciences* 8 (1972):237–242; D. E. Leary, “Immanuel Kant and the development of modern psychology,” in W. R. Woodward and M. G. Ash, eds., *The Problematic Science: Psychology in Nineteenth Century Thought* (New York: Praeger, 1982), pp. 17–42; and G. Verwey, *Psychiatry in an Anthropological and Biomedical Context* (Dordrecht: D. Reidel, 1985).

8 Kant’s *Anthropologie in pragmatischer Hinsicht* (1798) gives a good idea of what such a psychology would look like. See the translation *Anthropology from a Pragmatic Point of View*, by M. J. Gregor (The Hague: Martinus Nijhoff, 1974).

9 See B. D. Mackenzie and S. L. Mackenzie, “The case for a revised systematic approach to the history of psychology,” *Journal of the History of the Behavioral Sciences* 10 (1974):324–347. The argument is based on E. A. Burt, *The Metaphysical Foundations of Modern Physical Science* (London: Routledge & Kegan Paul, 1932), and A. Koyré, *From the Closed World to the Infinite Universe* (Baltimore: Johns Hopkins University Press, 1957).

10 Edmund Husserl gives a succinct description of this process in his *The Crisis of European Sciences and Transcendental Phenomenology*, pt. 2 (Evanston, Ill.: Northwestern University Press, 1970).

11 See Rorty, *Mirror of Nature*.

12 J. F. Herbart, *Psychologie als Wissenschaft, neu gegründet auf Erfahrung, Metaphysik und Mathematik* (Königsberg: Unger, 1824); also, *Ueber die Möglichkeit und Nothwendigkeit, Mathematik auf Psychologie anzuwenden*, vol. 5 of *Sämtliche Werke*, ed. K. Kehrbach (Langensalza: Beyer, 1890).

13 Kant, *Anthropology*, p. 22.

14 The major German representatives of this latter trend were, in chronological order, J. F. Fries, F. E. Beneke, and A. Fortlage. See also D. E. Leary, "The philosophical development of the conception of psychology in Germany 1780–1880," *Journal of the History of the Behavioral Sciences* 14 (1978):113–121. None of these figures received much recognition from the dominant German philosophical circles, and Beneke suffered political persecution for his views.

15 In France the philosophical accents were somewhat different, but in Auguste Comte one meets the same combination of distrust of introspection and affirmation of supraindividual principles of order.

16 This was of course the prevailing orthodoxy in nineteenth-century British empiricist philosophy, in the Scottish school of philosophy, and in early American psychology. One reason for the revolutionary impact of behaviorism in the American context must certainly be traced to its provocative break with the received basis of Anglo-Saxon mental philosophy.

17 After about 1860 the attack on introspection as a method is typically grounded in a mixture of traditional philosophical arguments and more modern concerns for the claims of a supraindividual order. Leading examples of the new trend are the German neo-Kantian socialist F. A. Lange (see his *The History of Materialism*, vol. 2[1865], chap. 3), and the spokesman of British institutional psychiatry, Henry Maudsley (see his *The Physiology of Mind* [1867], chap. 1). Lange coined the slogan "psychology without the soul." Both he and Maudsley played an important role in popularizing the idea of a scientific psychology in the years preceding the establishment of Wundt's laboratory at Leipzig. On Lange's interesting biography, see O. A. Ellissen, *Friedrich Albert Lange: Eine Lebensbeschreibung* (Leipzig: Baedeker, 1891).

18 See D. Bloor, *Wittgenstein: A Social Theory of Knowledge* (New York: Columbia University Press, 1983), chap. 4.

19 Hacking, for instance, distinguishes between the test, the adventure, the diagnosis, and the dissection. See I. Hacking, *The Emergence of Probability* (Cambridge: Cambridge University Press, 1975), pp. 35–37.

20 W. Wundt, *Grundzüge der physiologischen Psychologie* (Leipzig: Engelmann, 1874). In the introduction to this crucial work Wundt defines the new approach as follows: "Psychological introspection goes hand in hand with the methods of experimental physiology. If one wants to put the main emphasis on the characteristic of the method, our science, experimental psychology, is to be distinguished from the ordinary mental philosophy [*Seelenlehre*] based purely on introspection" (pp. 2–3).

21 Because English physiology was still a few years away from this development, there could be no question of it nurturing the emergence of an experimental psychology at that time. See G. L. Geison, "Social and institutional factors in the

stagnancy of English physiology, 1840–1870,” *Bulletin of the History of Medicine* 46 (1972):30–58.

22 I speak of the hopes of the young Wundt advisedly because in due course he lost the enthusiasm for the possibilities of the experimental method in psychology that he had expressed in his first major publication. See the introduction to his *Beiträge zur Theorie der Sinneswahrnehmung* (Leipzig and Heidelberg: Winter, 1862).

23 J. M. D. Olmsted, *François Magendie* (New York: Schuman’s, 1944); O. Temkin, “Basic science, medicine and the romantic era,” *Bulletin of the History of Medicine* 37 (1963):97–129 (p. 121); see also P. F. Crane-field, *The Way in and the Way out: François Magendie, Charles Bell and the Roots of the Spinal Nerves* (Mt. Kisco, N.Y.: Futura, 1974).

24 There are some interesting parallels between the history of the institutional relationship of anatomy and physiology and of philosophy and psychology. Both relationships were transformed when the junior partner redefined its subject matter in terms of the kind of knowledge yielded by the practice of experimentation.

25 See J. Schiller, “Physiology’s struggle for independence in the first half of the nineteenth century,” *History of Science* 7 (1968):64–89.

26 For a detailed analysis of these changes, and the connection between functionalism and experimentalism, see W. R. Albury, “Experiment and explanation in the physiology of Bichat and Magendie,” *Studies in History of Biology* 1 (1977):47–132; a different view of the relation between Bichat and Magendie is presented in J. E. Lesch, *Science and Medicine in France* (Cambridge, Mass.: Harvard University Press, 1984). On the significance of the functional perspective, see also M. Foucault, *The Order of Things* (London: Tavistock, 1970), pp. 228 and 264.

27 See W. Coleman, “The cognitive basis of the discipline: Claude Bernard on physiology,” *Isis* 76 (1985):49–70.

28 I have analyzed this development more fully elsewhere. See K. Danziger, “Origins of the schema of stimulated motion: Towards a prehistory of modern psychology,” *History of Science* 21 (1983):183–210.

29 This received its most influential textbook codification in Johannes Müller’s, *Handbuch der Physiologie des Menschen* (Coblenz: Holscher, 1833–1840). An English translation appeared in 1842. For slightly later English language developments in this direction, see K. Danziger, “Mid-nineteenth century British psychophysiology: A neglected chapter in the history of psychology,” in W. R. Woodward and M. G. Ash, eds., *The Problematic Science: Psychology in Nineteenth Century Thought* (New York: Praeger, 1982), pp. 119–146.

30 The practical effect was mediated by the Russian physiological school of Sechenov and Pavlov and never became part of the mainstream of western psychology.

31 The classical example is of course Fechner’s appropriation of E. H. Weber’s experiments in sensory physiology with the purpose of providing some empirical foundation for his metaphysical intuitions about the soul. See M. E. Marshall, “Physics, metaphysics, and Fechner’s psychophysics,” in Woodward and Ash, *The Problematic Science*, pp. 65–87. This line of thought had been made possible, among other things, by the fact that the new experimental physiology forced a relocation of sensory experience in the central parts of the nervous system, the same parts that were often regarded as the seat of the soul. See M. Gross, “The lessened locus of feelings: A transformation in French physiology in the early nineteenth century,” *Journal of the History of Biology* 12 (1979):231–271.

32 Wundt, *Grundzüge*.

33 From Wundt's own statistical summary of the experimental publications emanating from his laboratory during its first quarter century, it appears that three-quarters of all these studies were of this type. Moreover, most of the others fell into the latter part of this time period. W. Wundt, "Das Institut für experimentelle Psychologie," in *Die Institute und Seminare der philosophischen Fakultät an der Universität Leipzig* (Leipzig: Hirzel, 1909), pp. 118–133.

34 For an incisive analysis of the principles involved in the practices of this first community of scientific experimenters, see S. Shapin and S. Schaffer, *Leviathan and the Air-Pump: Hobbes, Boyle, and the Experimental Life* (Princeton: Princeton University Press, 1985).

35 See Bringmann et al., "Wundt's laboratory"; and Wundt, "Das Institut."

36 For comprehensive statistics on this expansion, see W. Lexis, *Die deutschen Universitäten* (Berlin: Asher, 1893); also J. Conrad, "Ergebnisse der deutschen Universitätsstatistik," *Jahrbücher für Nationalökonomie und Statistik* 32 (1906): 433–492. Wundt himself presents an interesting analysis of the local factors that favored the success of his venture at the University of Leipzig; see W. Wundt, *Festrede zur funfhundertjährigen Jubelfeier der Universität Leipzig* (Leipzig: Engelmann, 1909).

37 M. A. Tinker, "Wundt's doctorate students and their theses," in Bringmann and Tweney, *Wundt Studies*, pp. 269–279; also S. Fernberger, "Wundt's doctorate students," *Psychological Bulletin* 30 (1933):80–83.

38 In the United States Titchener's laboratory at Cornell was particularly important in this respect. See R. D. Tweney, "Programmatic research in experimental psychology: E. B. Titchener's laboratory investigations, 1891–1927," in M. G. Ash and W. R. Woodward, eds., *Psychology in Twentieth-Century Thought and Society* (Cambridge: Cambridge University Press, 1987), pp. 35–57.

39 I have adopted the use of the term "technology" in this context and the distinction between material, literary and social technologies from Shapin and Schaffer, *Leviathan*. These are analytical distinctions; in the real world the different technologies are closely interwoven.

40 For the history of these forms, see C. E. McClelland, *State, Society, and University in Germany, 1700–1914* (Cambridge: Cambridge University Press, 1980), pp. 174–181.

41 The University of Berlin illustrates the trend: In 1811, 72 percent of its budget went on salaries and 24 percent on institutes; by 1896 salaries accounted for only 31 percent and institutes for 53 percent; see F. Paulsen, *Die deutschen Universitäten und das Universitätsstudium* (Berlin: Asher, 1902).

42 With the growth of the institute the organization of the work of its members began to assume distinctly industrial features. In 1893 Wundt describes an organization with coordinated work teams, team leaders, and planned division of topics. By then the institute apparently had twenty-five members; see Wundt, "Psychophysik und experimentelle Psychologie," in Lexis, *Die deutschen Universitäten*, pp. 454–455.

43 The most important of these were the chronoscope and the kymograph. For recently republished illustrations of these and other pieces of apparatus used in Wundt's laboratory, see J. A. Popplestone and M. White McPherson, "The vitality of the Leipzig model of 1880–1910 in the United States in 1950–1980," in Bringmann and Tweney, *Wundt Studies*, pp. 226–257; also W. Traxel, H. Gundlach, and U. Zschuppe, "Zur Geschichte der apparativen Hilfsmittel der Psychologie," in R. Brickenkamp, ed., *Handbuch der apparativen Verfahren der Psychologie* (Göt-

tingen: Hogrefe, 1986), pp. 1–22. See also M. Borell, “Instrumentation and the rise of physiology,” *Science and Technology Studies* 5 (1987):53–62.

44 At the beginning of his experimental program Wundt declared that experimental methods would fail in psychology “whenever an intelligent consent [*verständnisvolles Eingehen*] to the intentions of the psychologist could not be assumed” (p. 208). W. Wundt, “Die Aufgaben der experimentellen Psychologie” (1882) in *Essays*, 2d ed. (Leipzig: Engelmann, 1906), pp. 187–212.

45 See M. Sokal, ed., *An Education in Psychology: James McKeen Cattell’s Journal and Letters from Germany and England 1880–1888* (Cambridge, Mass.: MIT Press, 1980).

46 See the papers by Cattell, Friedrich, Merkel, Starke, Tischer, and Trautscholdt in the first three volumes of the journal *Philosophische Studien*, 1883–1886.

47 This may be an example of the process of *typification* described by P. Berger and T. Luckmann, *The Social Construction of Reality* (Harmondsworth: Penguin, 1967).

48 J. Kollert, “Untersuchungen über den Zeitsinn,” *Philosophische Studien* 1 (1883):78–89; V. Estel, “Neue Versuche über den Zeitsinn,” *Philosophische Studien* 2 (1885):37–65.

49 E.g., J. Merkel, “Die zeitlichen Verhältnisse der Willensthätigkeit,” *Philosophische Studien* 2 (1885):73–127. The same lack of consistency can be observed in the earliest English-language experimental reports; cf. the papers by G. S. Hall in *Mind*, vols. 10 and 11, and *American Journal of Psychology*, vol. 1.

50 The equipment of Wundt’s laboratory was not only copied; an industrial offshoot of the laboratory supplied such equipment to adequately funded customers. Wundt’s mechanic, Ernst Zimmermann, established his own firm in 1887 and produced psychological apparatus in series. This development certainly assisted the standardization of psychological research on an international scale. Subsequently, the firm of C. H. Stoelting in Chicago fulfilled a rather similar function. See Traxel, Gundlach and Zschuppe, “Hilfsmittel der Psychologie.”

Chapter 3. Divergence of investigative practice: The repudiation of Wundt

1 W. Wundt, “Die Aufgaben der experimentellen Psychologie” (1882), in *Essays*, 2d ed. (Leipzig: Engelmann, 1906), pp. 187–212.

2 See Franz Brentano, *Psychologie vom empirischen Standpunkt*, vol. 1 (1874; repr. Hamburg: Meiner, 1955), chap. 2. Although he does not acknowledge any indebtedness, Wundt was clearly aware of Brentano’s distinction and may well have decided to adopt it. Of course, the essential point goes back to Kant.

3 It is most unfortunate that English-language references to Wundt’s position have so often failed to reproduce this basic distinction and have generally used the term “introspection” to cover both concepts indiscriminately. This of course makes it impossible to understand Wundt’s practice.

4 W. Wundt, “Selbstbeobachtung und innere Wahrnehmung,” *Philosophische Studien* 4 (1887):292–309. For a concise account of Wundt’s conception of the relation of introspection and experiment at the time of the founding of his laboratory, see also his *Methodenlehre* of 1883 (pp. 282ff.). (This was the second volume of *Logik* [Stuttgart: Enke, 1883].)

5 A major function of experimental arrangements was the production of *precise*

reports. These arrangements “force introspection to give an answer to a precisely put question;” see W. Wundt, “Hypnotismus und Suggestion,” *Philosophische Studien* 8 (1893):1–85 (p. 65).

6 See, e.g., W. Wundt, “Ueber psychologische Methoden,” *Philosophische Studien* 1 (1883):1–40.

7 Early versions of such exclusions are to be found in the first edition of Wundt’s *Vorlesungen über die Menschen- und Thierseele* (Leipzig: Voss, 1863). Subsequently, the theme is developed in successive editions of his *Logik*, in his writings on *Völkerpsychologie*, and in his “Ueber Ausfrageexperimente und über die Methoden zur Psychologie des Denkens,” *Psychologische Studien* 3 (1907):301–360.

8 For an overview, see K. Danziger, “Origins and basic principles of Wundt’s *Völkerpsychologie*,” *British Journal of Social Psychology* 22 (1983):303–313.

9 See the preface to the third edition of Wundt’s *Logik*.

10 Cf. W. van Hoorn and T. Verhave, “Wundt’s changing conceptions of a general and theoretical psychology,” in W. G. Bringmann and R. D. Tweney, eds., *Wundt Studies: A Centennial Collection* (Toronto: C. J. Hogrefe, 1980), pp. 71–113.

11 For a more detailed discussion of this topic, see K. Danziger, “Wundt’s psychological experiment in the light of his philosophy of science,” *Psychological Research* 42 (1980):109–122.

12 W. Wundt, *Grundzüge der physiologischen Psychologie*, 5th ed., vol. 3 (Leipzig: Engelmann, 1903), p. 703.

13 For a version that is available in English, see part 5 of Wundt’s *Outlines of Psychology* (1897), reprinted in R. W. Rieber, ed., *Wilhelm Wundt and the Making of a Scientific Psychology* (New York: Plenum, 1980), pp. 179–195; see also T. Mischel, “Wundt and the conceptual foundations of psychology,” *Philosophical and Phenomenological Research* 31 (1970):1–26.

14 Wundt neither was nor referred to himself as a “structuralist.” This term was invented by Titchener for the purpose of distinguishing his own approach from that of many of the American psychologists. Some of the fundamental differences between Titchener and Wundt are discussed in the next section. For an overview of Wundt’s position, see A. Blumenthal, “Wilhelm Wundt: Psychology as the propaedeutic science,” in C. E. Buxton, ed., *Points of View in the Modern History of Psychology* (New York: Academic Press, 1985), pp. 19–49; K. Danziger, “Wundt and the two traditions of psychology,” and “Wundt’s theory of behavior and volition,” in Rieber, *Wundt and Scientific Psychology*, pp. 73–115; W. R. Woodward, “Wundt’s program for the new psychology: Vicissitudes of experiment, theory and system,” in W. R. Woodward and M. G. Ash, eds., *The Problematic Science: Psychology in Nineteenth Century Thought* (New York: Praeger, 1980), pp. 167–197.

15 See F. K. Ringer, *The Decline of the German Mandarins* (Cambridge, Mass.: Harvard University Press, 1969).

16 Among the more important of these contributions were Wundt’s *Ethik* (1886) and his *System der Philosophie* (1889), both published at Leipzig by Engelmann.

17 Cf. A. Métraux, “Wilhelm Wundt und die Institutionalisierung der Psychologie,” *Psychologische Rundschau* 31 (1980):84–98.

18 Wundt’s work must be understood in terms of the complex and fluid relationships between philosophy and science (and philosophy and psychology) in nineteenth-century German academic life. For a good modern overview of some

of this background, see H. Schnädelbach, *German Philosophy 1831–1933* (Cambridge: Cambridge University Press, 1983).

19 On G. E. Müller, see A. L. Blumenthal, “Shaping a tradition: Experimentalism begins,” in Buxton, *Modern History of Psychology*, chap. 3, pp. 51–83.

20 For further discussion of this concept, see chapter 8.

21 W. Wundt, “Die Psychologie im Kampf ums Dasein” (1913), in his *Kleine Schriften*, vol. 3 (Stuttgart: Kröner, 1921), pp. 515–543.

22 In his 1877 review of the state of philosophy in Germany, written for English-speaking readers, Wundt presents statistics on the number of lecture series devoted to the various subfields of philosophy at all the German universities during the period 1874–1877. The number of lecture courses in psychology exceeded those in metaphysics by a factor of about three and those in ethics by a factor of about four. However, logic was slightly more popular than psychology. See W. Wundt, “Philosophy in Germany,” *Mind* 2 (1877):493–518.

23 Cf. Windelband’s caustic comment on the situation: “For a time things in Germany had almost got to the point where the ability to occupy a chair in philosophy was regarded as established once someone had learned to push electrical buttons in a methodical manner.” W. Windelband, *Die Philosophie im deutschen Geistesleben des 19. Jahrhunderts* (Tübingen: Mohr, 1909), p. 92.

24 For a comprehensive account of this process, see M. G. Ash, “Wilhelm Wundt and Oswald Külpe on the institutional status of psychology: An academic controversy in historical context,” in Bringmann and Tweney, *Wundt Studies*, pp. 396–421; and M. G. Ash, “Academic politics in the history of science: Experimental psychology in Germany, 1879–1941,” *Central European History* 13 (1981):255–286. W. Wirth, one of the post-Wundtian generation of German experimental psychologists who was fairly close to Wundt’s position, refers to being victimized by the “philosophic countercurrent” in his autobiographical statement in C. Murchison, *A History of Psychology in Autobiography*, vol. 3 (Worcester, Mass.: Clark University Press, 1936), pp. 283–327 (p. 292).

25 For further discussion of these differences, see chapter 8, and K. Danziger, “The social origins of modern psychology,” in A. R. Buss, ed., *Psychology in Social Context* (New York: Irvington, 1979), pp. 27–45.

26 For a more detailed analysis of this development, see K. Danziger, “The positivist repudiation of Wundt,” *Journal of the History of the Behavioral Sciences* 15 (1979):205–230.

27 The major philosophical underpinning for these positions were provided by Ernst Mach and R. H. L. Avenarius whose writings exerted a strong influence on some of the key figures among the generation of experimental psychologists that immediately followed Wundt. The latter attacked this position in a series of long papers in *Philosophische Studien* between 1894 and 1898. For details, see Danziger, “The Positivist Repudiation of Wundt.”

28 E. B. Titchener, “Prolegomena to a study of introspection,” *American Journal of Psychology* 23 (1912):427–448 (p. 427). See also Titchener, “The problems of experimental psychology,” *American Journal of Psychology* 16 (1905):208–224.

29 Titchener himself, Alfred Binet, and Theodor Lipps should probably be counted as the most prominent non-Würzburg proponents of a less restrictive use of introspection in an experimental context. On the Würzburg experiments, see G. Humphrey, *Thinking* (London: Methuen, 1950).

30 W. Wundt, “Ueber Ausfrageexperimente und über die Methoden zur Psychologie des Denkens,” *Psychologische Studien* 3 (1907):301–360.

31 My account of these changes was originally published in K. Danziger, "The history of introspection reconsidered," *Journal of the History of the Behavioral Sciences* 16 (1980):241–262.

32 The most explicit advocate of these methods was Narziss Ach, who coined the term "systematic experimental introspection." See N. Ach, *Ueber die Willenstätigkeit und das Denken* (Göttingen: Vandenhoeck und Ruprecht, 1905).

33 Cf. Ach, *ibid.*, and J. Segal, "Ueber den Reproduktionstypus und das Reproduzieren von Vorstellungen," *Archiv für die gesamte Psychologie* 12 (1908):124–235.

34 G. E. Müller, *Zur Analyse der Gedächtnistätigkeit und des Vorstellungsverlaufes* (1911; repr., Leipzig: Barth, 1924).

35 F. Galton, *Inquiries into Human Faculty and Its Development* (London: Macmillan, 1883).

36 A. Binet. "La mesure de la sensibilité," *L'Année psychologique* 9 (1903):79–128.

37 These studies are discussed in more detail in chapter 9 of the present volume.

38 See Shapin and Schaffer, *Leviathan*.

39 The plea for the reorientation of psychological research toward the production of knowledge useful for human management was central to Watson's behaviorist manifesto: "If psychology would follow the plan I suggest the educator, the physician, the jurist and the businessman could utilize our data in a practical way, as soon as we are able, experimentally, to obtain them." J. B. Watson, "Psychology as the behaviorist views it," *Psychological Review* 20 (1913):158–177. Note the stark appeal for a change of disciplinary project: "If psychology would follow the plan I suggest."

40 Although Titchener would not have wished to equate the viewpoint of psychology with that of physical science, his strictures on the language of introspection represented a kind of *reductio ad absurdum* of the neutral, value-free, unambiguous language of scientific description. He insisted on the elimination, not only of value and ambiguity, but of meaning itself. The classical statement of his position is to be found in E. B. Titchener, "Description vs. statement of meaning," *American Journal of Psychology* 23 (1912):165–182; also see his "The schema of introspection" in the same volume, pp. 485–508.

41 E. von Aster, "Die psychologische Beobachtung und experimentelle Untersuchung von Denkvorgängen," *Zeitschrift für Psychologie* 49 (1908):56–107.

42 Essentially the same distinction as that made by von Aster was also made independently, though less elegantly, by Dürr, a competent experimenter who had himself been a subject in Bühler's Würzburg experiments. See E. Dürr, "Ueber die experimentelle Untersuchung der Denkvorgänge," *Zeitschrift für Psychologie* 49 (1908):315. Titchener accepted his own version of the distinction in which the place of *Kundgabe* is taken by "information." Characteristically, this was an intellectualistic version of the original distinction in which description is now contrasted with interpretation and not with expression, as was the original intention. As usual, the spirit of Titchener survived in his student; see E. G. Boring's treatment of the issue in *A History of Experimental Psychology* (New York: Appleton-Century-Crofts, 1950), p. 610. After a gap of several decades there has been a more recent revival of interest in the topic of introspection. See W. Lyons, *The Disappearance of Introspection* (Cambridge, Mass.: MIT Press, 1986).

43 Von Aster, "Die psychologische Beobachtung."

Chapter 4. The social structure of psychological experimentation

1 R. M. Farr, "Social representations: Their role in the design and execution of laboratory experiments," in R. M. Farr and S. Moscovici, eds., *Social Representations* (Cambridge: Cambridge University Press, 1984).

2 M. Mehner, "Zur Lehre vom Zeitsinn," *Philosophische Studien* 2 (1885):546–602.

3 For example, Wundt appears as subject in the following papers published by his students in the first volume of the journal *Philosophische Studien* (1883–1884): M. Friedrich, "Ueber die Apperceptionsdauer bei einfachen und zusammengesetzten Vorstellungen," pp. 39–77; J. Kollert, "Untersuchungen über den Zeitsinn," pp. 78–89; M. Trautscholdt, "Experimentelle Untersuchungen über die Association der Vorstellungen," pp. 213–250; E. Tischer, "Ueber die Unterscheidung von Schallstärken," pp. 495–542.

4 The case of two of Wundt's students, Cattell and Berger, has been documented in some detail on the basis of Cattell's letters in M. Sokal, ed., *An Education in Psychology: James McKeen Cattell's Journals and Letters from Germany and England, 1880–1888* (Cambridge, Mass.: MIT Press, 1980). Another early instance of a pair of investigators who alternated as experimenter and subject is provided by Lorenz and Merkel; see G. Lorenz, "Die Methode der richtigen und falschen Fälle in ihrer Anwendung auf Schallempfindungen," *Philosophische Studien* 2 (1885):394–474.

5 See, e.g., L. Lange, "Neue Experimente über den Vorgang der einfachen Reaction auf Sinneseindrücke," *Philosophische Studien* 4 (1888):497–510; and E. B. Titchener, "Zur Chronometrie des Erkennungsactes," *Philosophische Studien* 8 (1893):138–144.

6 For a succinct insider's account of the relevant institutional practices, see the translation of a late nineteenth-century German guidebook, F. Paulsen, *The German Universities and University Study* (New York: Scribner, 1906), p. 212: "The student can be introduced to scientific research only by the method of cooperation. And that is the real purpose of the seminars; they are the nurseries of scientific research. In them, under the guidance and assistance of a master, pupils become acquainted with scientific work and learn how to do it. After their apprenticeship they continue to work themselves as masters." On the significance of the German university system for the organization of scientific work, see R. Whitley, *The Intellectual and Social Organization of the Sciences* (Oxford: Clarendon Press, 1984).

7 C. Richet, "De l'influence des mouvements sur les idées," *Revue philosophique* 8 (1879):610–615; C. Richet, "Du somnambulisme provoqué," *Revue philosophique* 10 (1880):337–374, 462–493; H. Beaunis, "L'expérimentation en psychologie par le somnambulisme provoqué," *Revue philosophique* 20 (1885):1–36; H. Beaunis, "Etudes physiologiques et psychologiques sur le somnambulisme provoqué" (Paris: Alcan, 1886); A. Binet and C. Féré, "Le hypnotisme chez les hystériques: Le transfert psychique," *Revue philosophique* 19 (1885):1–25; J. Delboeuf, "La mémoire chez les hypnotisés," *Revue philosophique* 21 (1886):441–472; J. Delboeuf, "De l'influence de l'éducation et de l'imitation dans le somnambulisme provoqué," *Revue philosophique* 22 (1886):146–171. My account of this work was originally published as part of a paper on "The origins of the psychological experiment as a social institution," *American Psychologist* 40 (1985):133–140.

8 The social structure of the clinical experiment was much closer to that of most contemporary psychological experiments than was the Leipzig experiment. In this connection it is interesting to note that the period of concern with the social psychology of the psychological experiment was ushered in by an analogy between role playing in hypnosis and role playing in experimental situations. See M. T. Orne, "On the social psychology of the psychological experiment: With particular reference to demand characteristics and their implications," *American Psychologist* 17 (1962):776–783; also M. T. Orne, "Hypnosis, motivation, and the ecological validity of the psychological experiment," *Nebraska Symposium on Motivation* 18 (1970):187–265.

9 A. Binet, "Recherches sur les mouvements chez quelques jeunes enfants," *Revue philosophique* 29 (1890):297–309.

10 *Grand Larousse de la langue française*, vol. 6 (Paris: Librairie Larousse, 1973); E. Littré, *Dictionnaire de la langue française*, vol. 7 (Paris: Gallimard-Hachette 1968).

11 E.g., in C. Féré, "Sensation et mouvement," *Revue philosophique* 20 (1885):337–368. This usage is also found in the titles of publications, e.g., by Bottey, "Des suggestions provoquées à l'état de veille chez les hystériques et chez les sujet sains" (1884), and by Brémaud, "Hypnotisme chez des sujet sains" (1883), as cited in M. Dessoir, *Bibliographie des Hypnotismus* (Berlin: Duncker, 1888).

12 *Oxford English Dictionary*, vol. 10 (Oxford: Clarendon Press, 1933). The term crops up occasionally in James Braid's *Neurypnology* of 1843; see *Braid on Hypnotism: The Beginnings of Modern Hypnosis* (New York: Julian Press, 1960), p. 209. In later years psychic researchers sometimes investigated hypnotic phenomena and used the term, probably as part of an attempt to establish their scientific credentials. See C. S. Alvarado, "Note on the use of the term *Subject* in pre-1886 discussions of thought transference," *American Psychologist* 42 (1987):101–102.

13 G. S. Hall, "Reaction time and attention in the hypnotic state," *Mind* 8 (1883):170–182; E. Gurney, "The stages of hypnotism," *Mind* 9 (1884):110–121.

14 G. S. Hall and H. H. Donaldson, "Motor sensations on the skin," *Mind* 10 (1885):557–572; G. S. Hall and J. Jastrow, "Studies of rhythm," *Mind* 11 (1886):55–62.

15 J. M. Cattell, "The time taken up by cerebral operations," *Mind* 11 (1886):220–242; J. M. Cattell and S. Bryant, "Mental association investigated by experiment," *Mind* 14 (1889):230–250.

16 See E. Gurney, "Further problems of hypnotism," *Mind* 12 (1887):112–222; E. Gurney, F. W. H. Myers, and F. Podmore, *Phantoms of the Living* (London: Trubner, 1886); J. Jacobs, "Experiments on 'prehension,'" *Mind* 12 (1887):75–79.

17 The anthropometric laboratory subsequently reopened at the Science Museum, South Kensington. For an account of these laboratories and Galton's use of the knowledge they produced, see K. Pearson, *The Life, Letters and Labours of Francis Galton*, vol. 2 (Cambridge: Cambridge University Press, 1924), pp. 357–359; 370–386; also F. Galton, *Memories of My Life* (London: Methuen, 1908) pp. 244–250.

18 L. S. Hearnshaw, *A Short History of British Psychology 1840–1940* (London: Methuen, 1964), p. 19. D. W. Forrest, *Francis Galton: The Life and Work of a Victorian Genius* (London: Paul Elk, 1974), p. 37; K. Pearson, *The Life, Letters and Labours of Francis Galton*, vol. 1 (Cambridge: Cambridge University Press, 1914), p. 157. On the significance of phrenology for some of Galton's cohorts, see

R. M. Young, *Mind, Brain and Adaptation in the Nineteenth Century* (Cambridge: Cambridge University Press, 1970).

19 Galton was notoriously wrong about the general importance of the performances he measured – such matters as visual acuity and strength of hand grip – but this does not of course affect the significance of his research goals. Those who followed in his footsteps eventually came up with measures of less trivial performances; see R. Fancher, *The Intelligence Men* (New York: Norton, 1985).

20 R. Cooter, *The Cultural Meaning of Popular Science: Phrenology and the Organization of Consent in 19th Century Britain* (Cambridge: Cambridge University Press, 1984). On the relationship between the specialization of occupations and the emerging interest in the assessment of “individual differences,” see also A. R. Buss, “Galton and the birth of differential psychology and eugenics: Social, political and economic forces,” *Journal of the History of the Behavioral Sciences* (1976):12, 47–58.

21 A contractual model to describe the social situation in all types of psychological experiments has been proposed by H. Schuler, *Ethical Problems in Psychological Research* (New York: Academic Press, 1982). Although all experimental situations have their contractual aspects, the implication that all such situations are essentially contractual in nature is unfortunate. Such an approach may blind one to the possibilities of divergent types of experimental situations within which the contractual aspects may vary greatly in importance and significance. Moreover, these different types of experimental situations may entail different ethical considerations, and what is ethically appropriate in one context may not be so in another.

22 This terminology has been common since R. Rosenthal and R. L. Rosnow, eds., *Artifact in Behavioral Research* (New York: Academic Press, 1969). For a broader view on “artifacts,” see R. M. Farr, “On the social significance of artifacts in experimenting,” *British Journal of Social and Clinical Psychology* 17 (1978): 299–306.

23 These matters have been extensively explained in the work of C. Argyris; see, e.g., C. Argyris, “Some unintended consequences of rigorous research,” *Psychological Bulletin* 70 (1969):185–197.

24 “Because of Charcot’s paternalistic attitude and his despotic treatment of students, his staff never dared contradict him; they therefore showed him what they believed he wanted to see. After rehearsing the demonstrations, they showed the subjects to Charcot, who was careless enough to discuss their cases in the patients’ presence. A peculiar atmosphere of mutual suggestion developed between Charcot, his collaborators, and his patients, which would certainly be worthy of an accurate sociological analysis.” It would indeed. H. F. Ellenberger, *The Discovery of the Unconscious* (New York: Basic Books, 1970), p. 98.

25 For an analysis of the experiments by Titchener’s students, written long after the dust had settled, see G. Humphrey, *Thinking: An Introduction to Its Experimental Psychology* (London: Methuen, 1950), pp. 119–129.

26 In the Leipzig mold: G. S. Hall and Y. Motora, “Dermal sensitiveness to gradual pressure changes,” *American Journal of Psychology* 1 (1887):72–98; in the tradition of experimental hypnosis studies: G. S. Hall, “Reaction time.” Those of Hall’s studies that were closer to Galtonian practices will be discussed in the next chapter; they were the innovative ones.

27 Cattell contributed the term “mental test”: J. M. Cattell, “Mental tests and measurements,” *Mind* 15 (1890):373–381; see also J. M. Cattell and L. Farrand,

"Physical and mental measurements of the students of Columbia University," *Psychological Review* 3 (1896):618–648.

28 For a comprehensive overview, see J. A. Popplestone and M. White McPherson, "Pioneer psychology laboratories in clinical settings," in J. Brozek, ed., *Explanations in the History of Psychology in the United States* (Lewisburg, Penn.: Bucknell University Press, 1984), pp. 196–272.

29 E. L. Thorndike, *Animal Intelligence* (New York: Macmillan, 1911), p. 3.

30 For the period 1934–1936 I was only able to locate three American studies involving an exchange of experimenter and subject roles that were not in the area of sensation and perception, although in the previous decade the practice is still in wider use.

Chapter 5. The triumph of the aggregate

1 J. Ziman, *Public Knowledge: The Social Dimensions of Science* (Cambridge: Cambridge University Press, 1968).

2 S. Woolgar, "Discovery: Logic and sequence in a scientific text," and N. Gilbert and M. Mulkay, "Contexts of scientific discourse: Social accounting in experimental papers," both in K. D. Knorr, R. Krohn, and R. Whitley, eds., *The Social Process of Scientific Investigation* (Dordrecht: Reidel, 1980), pp. 239–268 and 269–294. See also R. Whitley, "Knowledge producers and knowledge acquirers," in T. Shinn and R. Whitley, eds., *Expository Science: Forms and Functions of Popularization* (Dordrecht: D. Reidel, 1985), pp. 3–28.

3 K. D. Knorr-Cetina, *The Manufacture of Knowledge* (New York: Pergamon, 1981).

4 This divergence is analyzed further in K. Danziger, "The positivist repudiation of Wundt," *Journal of the History of the Behavioral Sciences* 15 (1979):205–230.

5 In the case of Claude Bernard, probably the most celebrated exponent of this approach, advocacy of its virtues was coupled with derision for the statistical approach: "We shall therefore try to find these experimental, physiological conditions, instead of tabulating the variations in phenomena and taking averages as expressions of reality; we should thus reach conclusions based on correct statistics, but with no more scientific reality than if they were wholly arbitrary." Claude Bernard, *An Introduction to the Study of Experimental Medicine* (1865; repr. New York: Dover, 1957), pp. 116ff.

6 For Pavlov's investigative practice, see his *Conditioned Reflexes: An Investigation of the Physiological Activity of the Cerebral Cortex*, trans. and ed. G. V. Anrep (London: Oxford University Press, 1927).

7 An instructive example of this conception of psychological experimentation is provided by Titchener's defense of classical reaction-time studies against the criticism of Baldwin who preferred a typological interpretation of individual differences observed in such situations. See E. B. Titchener, "The type theory of simple reactions," *Mind* n.s. 4 (1895):506–514. The fundamental implications of the Baldwin–Titchener controversy have been noted more recently by D. L. Krantz, in *Schools of Psychology* (New York: Appleton Century Crofts, 1965), and by Gernot Böhme, "Cognitive norms, knowledge interests and the constitution of the scientific object: A case study in the functioning of rules for experimentation," in E. Mendelsohn, P. Weingart, and R. Whitley, eds., *The Social Production of Scientific Knowledge* (Dordrecht: Reidel, 1977). As representatives of major historical orientations within psychology, Baldwin and Titchener were not on the same

level, however. Whereas Titchener gave highly coherent expression to a well-established position, then at the height of its development, Baldwin represented a new viewpoint whose methodological implications had not been fully worked out.

8 K. H. Metz, "Paupers and numbers: The statistical argument for social reform in Britain during the period of industrialization," in L. Krüger, L. J. Daston, and M. Heidelberger, eds., *The Probabilistic Revolution*, vol. 1: *Ideas in History* (Cambridge, Mass.: MIT Press, 1987), pp. 337–350; and T. M. Porter, *The Rise of Statistical Thinking 1820–1900* (Princeton: Princeton University Press, 1986), chap. 1.

9 For some of the historical background on questionnaires, see R. H. Gault, "A history of the questionnaire method of research in psychology," *Pedagogical Seminary* 14 (1907):366–383.

10 S. Jaeger, "Volkspsychologie, Statistik und Sozialreform," in S. Bem, H. Rappard, and W. van Hoorn, eds., *Studies in the History of Psychology and the Social Sciences*, vol. 2 (Leiden: Psychologisch Instituut Rijksuniversiteit Leiden, 1984), pp. 148–160.

11 G. S. Hall, "Contents of children's minds on entering school," *Pedagogical Seminary* 1 (1891):139–173.

12 B. Sigismund, *Kind und Welt* (Braunschweig: Vieweg, 1856). Nineteenth-century medical literature played an important role in establishing a place for statistics in scientific rhetoric. See J. H. Cassedy, *American Medicine and Scientific Thinking 1800–1860* (Cambridge, Mass.: Harvard University Press, 1986).

13 C. Darwin, *The Expression of the Emotions in Man and Animals* (1872; repr. Chicago: University of Chicago Press, 1965); F. Galton, *English Men of Science* (1874; repr. London: Frank Cass, 1970), and *Inquiries into Human Faculty* (London: Macmillan, 1883).

14 The pages of Hall's *Pedagogical Seminary* contain many studies of this type. For example, in the third volume of 1895–1896 we find: J. A. Jancock, "A preliminary study of motor ability in children" – 158 subjects; M. Schallenberger, "A study of children's rights as seen by themselves" – over 3,000 subjects; M. C. Holmes, "The fatigue of a school hour" – 150 subjects; E. Barnes, "Punishment as seen by children" – 4,000 subjects; K. Fackenthal, "The emotional life of children" – 200 subjects; M. A. Herrick, "Children's drawings" – 450 subjects; H. C. Kratz, "Characteristics of the best teacher as recognized by children" – 2,400 subjects; H. P. Kennedy, "Effect of high-school work upon girls during adolescence" – 129 subjects.

15 See Porter, *Statistical Thinking*, pp. 57–70. In his earliest statement on the methods of psychology, the introduction to the *Beiträge zur Theorie der Sinneswahrnehmung* (Leipzig and Heidelberg: C. F. Winter, 1862), Wundt mentions both statistics and experiment as the methods for placing psychology on a scientific footing. However, by the time his laboratory had begun operating in Leipzig, statistics had been very much downgraded compared to experiment. See, e.g., his *Methodenlehre* of 1883, p. 494 (W. Wundt, *Logik*, vol. 2 [Stuttgart: Enke, 1883]).

16 L. A. Quetelet, *Sur l'homme et le développement de ses facultés* (Paris: Bachelier, 1835); Eng. trans., *A Treatise on Man and the Development of His Faculties* (Gainesville, Fla.: Scholars' Facsimiles, 1969).

17 S. P. Turner, *The Search for a Methodology of Social Science* (Dordrecht: Reidel, 1986), chap. 4. See also G. Gigerenzer et al., *The Empire of Chance* (Cambridge: Cambridge University Press, 1989), chap. 2.

18 Porter, *Statistical Thinking*, chap. 6; and the related discussion in I. Hacking,

"Prussian numbers 1860–1882," in L. Krüger, L. J. Daston, and M. Heidelberger, eds., *The Probabilistic Revolution*, vol. 1: *Ideas in History* (Cambridge, Mass.: MIT Press, 1987), pp. 377–394.

19 H. T. Buckle, *History of Civilization in England*, 2 vols. (London: J. W. Parker, 1858–1861).

20 On the fundamental distinction between "essentialism" and "population thinking" in biology, see E. Mayr, *The Growth of Biological Thought: Diversity, Evolution and Inheritance* (Cambridge, Mass.: Harvard University Press, 1982).

21 On the implications of the Galtonian basis for psychology, and especially for its role in society, see N. Rose, *The Psychological Complex: Psychology, Politics and Society in England 1869–1939* (London: Routledge & Kegan Paul, 1985).

22 The distinction was quite clear to some nineteenth-century statisticians. F. Y. Edgeworth, for instance, expressed it in terms of a distinction between "observations" and "statistics": "Observations and statistics agree in being grouped about a Mean; they differ, in that the Mean of observations is real, of statistics is fictitious. The mean of observations is a cause . . . the mean of statistics is a description, a representative quantity put for a whole group. . . . Different measurements of the same man are observations; but measurements of different men, grouped round l'homme moyen, are prima facie at least statistics." Edgeworth, 1885, cited in S. M. Stigler, *The History of Statistics: The Measurement of Uncertainty before 1900* (Cambridge, Mass.: Harvard University Press, 1986), p. 309.

23 One still meets this approach in the well-known twentieth-century work of Hall's student, Arnold Gesell; see, e.g., A. Gesell et al., *First Five Years of Life: A Guide to the Study of the Preschool Child* (New York: Harper, 1940); or A. Gesell et al., *The Child from Five to Ten* (New York: Harper, 1946).

24 Explicit comments on the shortcomings of the "statistical" approach to psychological questions were common in the European literature – e.g., Binet in *L'étude expérimentale de l'intelligence* (Paris: Schleicher, 1903): "The Americans who love to do things big, often publish experiments made on hundreds and even thousands of persons. They believe that the conclusive value of a work is proportional to the number of observations made. This is an illusion. . . . If I have been able to throw some light by the attentive study of two subjects, it is because I have seen their behavior from day to day and have probed it over a period of several years. . . . We should prefer experiments that we can make on persons whose character and way of life are familiar to us" (Eng. trans. from T. H. Wolf, *Alfred Binet* [Chicago: University of Chicago Press, 1973], p. 120). Some samples of German criticisms of large samples will be found in H. Münsterberg, *Grundzüge der Psychologie* (Leipzig: Barth, 1900); C. Stumpf, "Zur Methodik der Kinderpsychologie," *Zeitschrift für pädagogische Psychologie und Pathologie* 2 (1900):17–21; W. Ament, "Logik der statistischen Methode," introduction to *Die Entwicklung der Pflanzenkenntnis beim Kinde und bei Völkern* (Berlin: Reuther and Reichard, 1904), pp. 3–14. The last-named author sums up a lengthy discussion with the epigrammatic statement (p. 14), "Die Masse aber trübt dem Forscher den Blick" (the mass blurs the investigator's gaze).

25 An uncritical acceptance of "official" categories like crime, suicide, etc., had of course been the basis for the application of statistics to social phenomena, starting with Quetelet. Now, the same naive acceptance of conventional categories was transferred to psychology. In both cases the investigators were able to reap the immediate advantages of using the same conceptual network as the one in which ordinary people organized their experience and expressed their concerns. The apparent practical relevance of such studies seemed obvious, whereas psychological

experiments deconstructed common experience, often without reestablishing links with it. Today, there is much greater awareness of the problematic nature of the social categories that form the basis both of social statistics and of experiments on social conduct; e.g., J. Irvine, I. Miles, and J. Evans, eds., *Demystifying Social Statistics* (London: Pluto, 1979); and K. J. Gergen, *Toward Transformation in Social Knowledge* (New York: Springer, 1982).

26 “The school is itself a kind of intelligence test, and the eventual modernization of American education meant the intensifying of actions that could produce estimates of learning ability.” D. Calhoun, *The Intelligence of a People* (Princeton: Princeton University Press, 1973), p. 76. See also Rose, *Psychological Complex*.

27 Even William James was not immune to the appeal of such a science, although it was left to others to convert his ideas into appropriate practices: “All natural sciences aim at practical prediction and control and in none of them is this more the case than in psychology to-day. We live surrounded by an enormous body of persons who are most definitely interested in the control of states of mind, and in incessantly craving for a sort of psychological science which will teach them how to act. What every educator, every asylum superintendent, asks of psychology is practical rules. Such men care little or nothing about the ultimate philosophic grounds of mental phenomena, but they do care immensely about improving the ideas, dispositions, and conduct of the particular individuals in their charge.” W. James, “A plea for psychology as a ‘natural science,’” *Philosophical Review* 1 (1892):146–153 (p. 148). The premise, “that individuals in the national population were best described, analyzed, and interpreted by reference to the group in the national population to which they ‘belonged,’ ” has been called “the fundamental methodological premise of American evolutionary natural and social science.” See H. Cravens, “Applied science and public policy,” in M. Sokal (ed.), *Psychological Testing and American Society* (New Brunswick, N.J.: Rutgers University Press, 1987), pp. 158–194 (p. 179).

28 Most of the prominent figures in the early years of modern American psychology had little respect for the tradition of painstaking laboratory work represented by figures like Fechner, Wundt, and Titchener, either because of its “microscopic” character (James) or because of its purely academic interest (Hall). Also, this kind of psychology contributed little or nothing to satisfying the broad social demands that the American tradition placed on psychology as a teaching subject; cf. F. C. French, “The place of experimental psychology in the undergraduate courses,” *Psychological Review* 5 (1898):510–512. The problems faced by esoteric laboratory work in a culture hostile to intellectual elitism were not unique to psychology, but also arose in other sciences. See G. H. Daniels, *Science in American History: A Social History* (New York: Knopf, 1971), pp. 281ff.

29 In the original coding five categories of data were distinguished: (a) studies based exclusively on individual data; (b) studies essentially based on individual data but including some summaries; (c) studies essentially based on group data but including occasional reference to individual cases for purposes of illustration; (d) studies based exclusively on group data; and (e) truly mixed studies where both kinds of data carried some weight. However, categories (b) and (c) were found to show the same pattern of variation as the extreme categories (a) and (d). The pair (a) and (b) and the pair (c) and (d) were therefore collapsed into two combined categories. Together with (e) this resulted in a threefold classification which represents the overall trend more economically. Agreement among raters on the inclusion of articles in these categories ranged from 91.6 percent for the overall “group data” category to 83.5 percent for the overall “individual data” category.

30 In the case of two of the journals the first period actually represented the late 1910s because they only began publication in 1916 or 1917, and because they missed a year or two due to the war.

31 Some pertinent considerations are to be found in the following: W. van Hoorn, "Psychology and the reign of technology," in S. Bem, H. Rappard, and W. van Hoorn, eds., *Studies in the History of Psychology and the Social Sciences*, vol. 1 (Leiden: Psychologisch Instituut, 1983), pp. 105–118; W. van Hoorn and T. Verhave, "Socio-economic factors and the roots of American psychology: 1865–1914," *Annals of the New York Academy of Sciences* 291 (1977):203–221.

32 In this context the term "laboratory" has a social rather than a physical definition; it is not constituted by the presence of apparatus but by an investigative situation.

33 The importance of World War I intelligence testing for the social impact of psychology as a discipline is well recognized. See F. Samelson, "World War I intelligence testing and the development of psychology," *Journal of the History of the Behavioral Sciences* 13 (1977):274–282; and F. Samelson, "Putting psychology on the map: Ideology and intelligence testing," in A. R. Buss, ed., *Psychology in Social Context* (New York: Irvington, 1979), pp. 103–168.

34 See D. Bakan, *On Method* (San Francisco: Jossey Bass, 1967), chap. 1.

35 The metaphorical use of "organism" in connection with statistically constituted entities goes back to Gustav Theodor Fechner. See his *Kollektivmasslehre* (Leipzig: Engelmann, 1897).

36 At this stage in the history of psychology such transfers were often made in the service of one or another ideological position. A very effective means of promoting the social standing of psychology as a discipline was to hitch its investigative practices to one or another preexisting social ideology. Thus, intelligence testers claimed to have found the measure that radical hereditarians needed to enhance the plausibility of their eugenicist program, and behaviorists claimed to have found the techniques on which the effectiveness of programs of social control depended. The fundamental features that these two positions had in common were probably far more important than their superficial differences. It has often been pointed out that they shared the same undemocratic goals – e.g., J. Harwood, "The IQ in History," *Social Studies of Science* 13 (1983):465–477; R. Marks, *The Idea of IQ* (Washington, D.C.: University Press of America, 1981); F. Samelson, "On the science and politics of the IQ," *Social Research* 42 (1975):467–488. They also shared a simplistic intellectual style that was inseparable from their very real political role in mobilizing support for broad social programs. The intimate involvement with such ideological positions produced a simplistic formulation of psychological issues that left its traces on the discipline for a long time to come.

Chapter 6. Identifying the subject in psychological research

1 For a comprehensive account of the shaping of this double thrust, see J. M. O'Donnell, *The Origins of Behaviorism: American Psychology, 1870–1920* (New York: New York University Press, 1985).

2 By 1938, the proportion of published American psychological research studies that were based on work done with animal subjects had reached 15 percent of the total. This proportion had increased steadily over four decades. J. S. Bruner and G. W. Allport, "Fifty years of change in American psychology," *Psychological Bulletin* 37 (1940):757–776 (p. 764). Moreover, those who worked with animal

subjects tended to show a higher research productivity than those who worked with human subjects – a likely index of high status in the discipline. S. W. Fernberger, “The scientific interests and scientific publications of the members of the American Psychological Association,” *Psychological Bulletin* 35 (1938):261–281. A quarter of the papers delivered at the 1939 annual meeting of the APA were based on animal research; see Bruner and Allport, *ibid.* p. 765.

3 The problem is not peculiar to scientific psychology; see K. D. Knorr-Cetina, “The ethnographic study of scientific work: Towards a constructivist interpretation of science,” in K. D. Knorr-Cetina and M. Mulkay, eds., *Science Observed: Perspectives on the Social Study of Science* (London and Beverly Hills, Calif.: Sage, 1983), pp. 115–140.

4 See Popplestone and McPherson, “Pioneer psychology laboratories.”

5 This feature is discussed in chapter 9.

6 As indicated in the previous chapter, this approach was particularly common in the area of child study, but it was not limited to this area. Early examples of psychological census taking among adults are: J. Jastrow, “Community and association of ideas: A statistical study,” *Psychological Review* 1 (1894):152–158; and M. W. Calkins, “Synaesthesia,” *American Journal of Psychology* 7 (1895):90–107.

7 See, e.g., R. Smart, “Subject selection bias in psychological research,” *Canadian Psychologist* 7 (1966):115–121; D. P. Schultz, “The human subject in psychological research,” *Psychological Bulletin* 72 (1969):214–228; K. L. Higbee and M. G. Wells, “Some research trends in social psychology during the 1960’s,” *American Psychologist* 27 (1972):963–966.

8 J. F. Dashiell, “Note on use of the term ‘observer,’ ” *Psychological Review* 36 (1929):550.

9 M. Bentley, “‘Observer’ and ‘subject,’ ” *American Journal of Psychology* 41 (1929):682.

10 J. F. Dashiell, “A reply to Professor Bentley,” *Psychological Review* 37 (1930):183–185.

11 The empirical studies published in the *Psychological Review* before it became a theoretical journal tended to have the same character as those published in the *American Journal of Psychology*, and later, in the *Journal of Experimental Psychology*. *Psychological Monographs*, however, showed considerable variation on the kinds of studies published, also from one time period to another.

12 This is particularly the case for the *Journal of Educational Psychology*.

Chapter 7. Marketable methods

1 A telling little interchange is recalled by Charles Judd who was to achieve a position of considerable prominence in American educational psychology. In 1898, shortly after his return from Wundt’s laboratory, he had been appointed to a professorship at the School of Pedagogy of New York University. The memory of the following incident seems to have remained with him ever after: “I recall very well that I had on one occasion been lecturing enthusiastically on Weber’s Law to a class of New York City teachers who were seeking increases in their salaries by listening to me, when I was interrupted by one of my gray-haired auditors with this question: ‘Professor, will you tell us how we can use this principle to improve our teaching of children?’ I remember that question better than I do my answer.” C. H. Judd in C. Murchison, ed., *A History of Psychology in Autobiography*, vol. 2 (Worcester, Mass: Clark University Press, 1932), pp. 221–222.

2 L. Zenderland, "Education, evangelism, and the origins of clinical psychology: The child-study legacy," *Journal of the History of the Behavioral Sciences* 24 (1988):152–165.

3 The rate of expansion of the American educational system had reached staggering proportions. Between 1890 and 1920 there was at least one public high school built for each day of the calendar year and the enrollment increase was of the order of 1,000 percent. Between 1902 and 1913 public expenditure on education more than doubled; between 1913 and 1922 it tripled. The relation between changes in American education and the requirements of the social system has often been explored; see, e.g., C. J. Karrier, ed., *Shaping the American Educational State* (New York: Free Press, 1975); S. Bowles and H. Gintis, *Schooling in Capitalist America* (New York: Basic Books, 1976); J. Spring, *Education and the Rise of the Corporate State* (Boston: Beacon Press, 1972).

4 A. G. Powell, "Speculations on the early impact of schools of education on educational psychology," *History of Education Quarterly* 11 (1971):406–412, who speaks of "the deeply felt need of careerist male educators to separate themselves as much as possible from classroom teaching. Teaching, by and large, was viewed as transient, unrewarding, unprofessional, and female. Separate organizations, journals, and university courses were demanded by former teachers seeking a better and more respectable career identity through specialized administrative positions" (p. 410). See also R. E. Callahan and H. W. Button, "Historical change and the role of the man in the organization: 1865–1950," in D. E. Griffiths, ed., "Behavioral Science and Educational Administration," *Yearbook, National Society for the Study of Education* 63 (1964):73–92. On the role of testing in the educational system of the time, see D. P. Resnick, "Educational policy and the applied historian," *Journal of Social History* 14 (1981):539–559.

5 F. Bobbitt, "Some general principles of management applied to the problems of city-school systems," *Yearbook, National Society for the Study of Education* 12 (1913):7–96 (p. 12).

6 For a sampling of such analogies, see R. E. Callahan, *The Cult of Efficiency* (Chicago: University of Chicago Press, 1962); and D. Tyack and E. Hansot, *Managers of Virtue: Public School Leadership in America, 1820–1980* (New York: Basic Books, 1982).

7 F. E. Spaulding, "The application of the principles of scientific management," (NEA *Proceedings*, 1913) as cited in Callahan, *Cult of Efficiency*, p. 68.

8 In the standard text on school administration, the role of research departments in school districts is described as follows: "to study the needs of life and the industries, with a view to restating the specifications for the manufacture of the educational output; to study means for increasing the rate of production, and for eliminating the large present waste in manufacture; . . . to test out different methods of procedure, and gradually to eliminate those which do not give good results." E. P. Cubberley, *Public School Administration* (Boston: Houghton-Mifflin, 1916), p. 336. See also G. D. Strayer, "Measuring results in education," *Journal of Educational Psychology* 2 (1911):3–10.

9 See G. Joncich, *The Sane Positivist: A Biography of Edward L. Thorndike* (Middletown, Conn.: Wesleyan University Press, 1968). On the liaison between American education and psychological research, see also E. V. Johanningsmeier, "American educational research: Applications and misapplications of psychology to education," in T. V. Smith and D. Hamilton, eds., *The Meritocratic Intellect: Studies in the History of Educational Research* (Aberdeen: Aberdeen University Press, 1980), pp. 41–57.

10 E. C. Sanford, "Experimental pedagogy and experimental psychology," *Journal of Educational Psychology* 1 (1910):590–595. See also C. Seashore, "The educational efficiency engineer," *Journal of Educational Psychology* 4 (1913):244. For other psychologists involved in this area, see R. I. Watson, "A brief history of educational psychology," *The Psychological Record* 11 (1961):209–242.

11 R. L. Church, "Educational psychology and social reform in the progressive era," *History of Education Quarterly* 11 (1971):390–405.

12 On the significance of the war work for the establishment of psychology as a socially relevant science, see F. Samelson, "Putting psychology on the map," pp. 103–168.

13 R. S. Woodworth, *Psychological Issues* (1920; repr. New York: Columbia University Press, 1939), p. 385. In an autobiographical statement, J. Jastrow, the first psychologist who was never anything else but a psychologist, writes: "To speak of the renaissance of psychology, especially in the American setting, without explicit recognition of the practical motive would be a glaring omission; for that renaissance found its momentum in the appeal to psychology for the regulation of human affairs. Applied psychology is in many a quarter the pay-vein that supports the mine. The educational application is the oldest and most comprehensive." C. Murchison, ed., *A History of Psychology in Autobiography*, vol. 1 (Worcester, Mass.: Clark University Press, 1930), p. 155. By 1910, three-quarters of those American psychologists who indicated an active interest in psychological applications expressed that interest in the field of education; see J. M. O'Donnell, *The Origins of Behaviorism: American Psychology, 1870–1920* (New York: New York University Press, 1985), p. 227.

14 There is an extensive literature in this area. See, e.g., L. Baritz, *The Servants of Power* (Middletown, Conn.: Wesleyan University Press, 1965); D. S. Napoli, *Architects of Adjustment: The History of the Psychological Profession in the United States* (Port Washington, N.Y.: Kennikat Press, 1981); R. Marks, *The Idea of IQ* (Washington, D.C.: University Press of America, 1981); J. Harwood, "The IQ in history," *Social Studies of Science* 13 (1983):465–477; M. M. Sokal, ed., *Psychological Testing and American Society* (New Brunswick, N.J.: Rutgers University Press, 1987). On the British case, see G. Sutherland, *Ability, Merit and Measurement* (Oxford: Clarendon Press, 1984).

15 E. L. Thorndike, "The contribution of psychology to education," *Journal of Educational Psychology* 1 (1910):5–12.

16 E. C. Sanford, "Methods of research in education," *Journal of Educational Psychology* 3 (1912):303–315.

17 L. J. Cronbach, "The two disciplines of scientific psychology," *American Psychologist* 12 (1957):671–684. Between 1921 and 1936 some five thousand articles on mental testing were published in various journals; see E. B. South, *An Index of Periodical Literature on Testing, 1921–1936* (New York: The Psychological Corporation, 1937).

18 See, e.g., A. Ciocco, "The background of the modern study of constitution," *Bulletin of the History of Medicine* 4 (1936):23–28; J. Graham, "Lavater's physiognomy in England," *Journal of the History of Ideas* 22 (1961):561–572.

19 See S. J. Gould, *The Mismeasure of Man* (New York: Norton, 1981).

20 A. Binet and V. Henri, "La psychologie individuelle," *L'Année psychologique* 2 (1895):411–465. See also S. E. Sharp, "Individual psychology: A study in psychological method," *American Journal of Psychology* 10 (1899):329–391; and T. H. Wolf, *Alfred Binet* (Chicago: University of Chicago Press, 1973).

21 W. Stern, *Die differentielle Psychologie in ihren methodischen Grundlagen* (Leipzig: Barth, 1911). Stern had published his first review of the literature on individual differences in 1900, several years before Galtonian mental tests came to dominate the field. His final position is represented by his *General Psychology from the Personalistic Standpoint* (New York: Macmillan, 1938). For a sympathetic American voice, see M. Bentley, "Individual psychology and psychological varieties," *American Journal of Psychology* 52 (1939):300–301.

22 J. M. Baldwin, "Types of reaction," *Psychological Review* 2 (1895):259–273; and "The type theory of reactions," *Mind* n.s. 5 (1896):81–90.

23 This assumption was usually expressed in the form of Thorndike's axiom that all qualitative differences between individuals are reducible to quantitative differences: "All intelligible differences are ultimately quantitative. The difference between any two individuals, if describable at all, is described by comparing the amounts which A possesses of various traits with amounts which B possesses of the same traits. In intellect and character, differences of kind between one individual and another turn out to be definable, if defined at all, as compound differences of degree." E. L. Thorndike, *Individuality* (Boston: Houghton-Mifflin, 1911), p. 5.

24 J. Valsiner, ed., *The Individual Subject and Scientific Psychology* (New York: Plenum, 1986).

25 For more extensive treatment of this topic, see N. Rose, *The Psychological Complex: Psychology, Politics and Society in England 1869–1939* (London: Routledge and Kegan Paul, 1985): "This psychology sought to establish itself by claiming its ability to deal with the problems posed for social apparatuses by dysfunctional conduct. . . . psychological normality was conceived of as . . . an absence of social inefficiency: that which did not need to be regulated" (pp. 5–6).

26 See R. Cooter, *The Cultural Meaning of Popular Science: Phrenology and the Organization of Consent in 19th Century Britain* (Cambridge: Cambridge University Press, 1984); and D. Bakan, "The influence of phrenology on American psychology," *Journal of the History of the Behavioral Sciences* 2 (1966):200–220.

27 The comprehensive history of the emergence of examinations as a major social institution in the nineteenth century remains to be written. Existing sources are widely scattered. The following provide access to material in this area: Rose, *Psychological Complex*; J. Roach, *Public-Examinations in England 1850–1900* (Cambridge: Cambridge University Press, 1971); R. J. Montgomery, *Examinations: An Account of their Evolution as Administrative Devices in England* (London: Longmans, 1965). Galton appears to have been fascinated by examinations and used results from the Indian Civil Service examinations in his attempts at developing a measure of statistical correlation. See R. E. Fancher, "The examined life: Competitive examinations in the thought of Francis Galton," *History of Psychology Newsletter* 16 (1984):13–20.

28 A contemporary reviewer presented the difference in practice as follows: "The French psychologists have tended on the whole to study the individual's reactions with a view to understanding the total adjustment of the organism or personality. The English and Americans, in large part, have concerned themselves with expressing the reactions in statistical terms. . . . Moreover, those who view intelligence testing merely as an aid to clinical diagnosis tend toward the older view, the French, while those who deal in mass materials would emphasize the latter and tend to define intelligence in statistical formulae"; see K. Young, "The history of mental testing," *The Pedagogical Seminary* 31 (1923):1–48 (p. 45). See also R. E. Fancher, *The Intelligence Men* (New York: Norton, 1985).

29 J. M. O'Donnell, "The clinical psychology of Lightner Witmer," *Journal of the History of the Behavioral Sciences* 15 (1979):3–17; B. Richards, "Lightner Witmer and the project of psychotechnology," *History of the Human Sciences* 1 (1988):201–219; J. A. Popplestone and M. W. McPherson, "Pioneer psychology laboratories in clinical settings," in J. Brozek, ed., *Explorations in the History of Psychology in the United States* (Lewisburg, Penn.: Bucknell University Press, 1984), pp. 196–272.

30 See F. Samelson, "Was early mental testing (a) racist inspired, (b) objective science, (c) a technology for democracy, (d) the origin of multiple-choice exams, (e) none of the above?" in Sokal, *Psychological Testing and American Society*, pp. 113–127.

31 Statisticians who did not share the Pearsonian world view were critical of what they considered a misuse of their craft by the mental-test movement. In his review of an early text on the statistics of mental testing, the British statistician Yule wrote: "Statistical methods . . . should be regarded as ancillary, not essential. They are essential where the subject of investigation is itself an aggregate, as a swarm of atoms or a crowd. But here the subject is the individual"; see G. Udny Yule, "Critical notice of Brown and Thomson's *Essentials of Mental Measurement*," *British Journal of Psychology* 12 (1921):105. On the fundamental differences between Yule and Pearson, see D. A. Mackenzie, *Statistics in Britain 1865–1930: The Social Construction of Scientific Knowledge* (Edinburgh: Edinburgh University Press, 1981).

32 R. S. Cowan, "Francis Galton's statistical ideas: The influence of eugenics," *Isis* 63 (1972):509–528; D. A. Mackenzie, *Statistics in Britain*.

33 It is more accurate to speak of a Galton–Pearson school, because while Galton provided the inspiration, Pearson provided the philosophically and technically complete formulation. However, the term "Galtonian" is more convenient as an adjectival form as long as it is understood as a kind of historical shorthand.

34 I previously presented this analysis in K. Danziger, "Statistical method and the historical development of research practice in American psychology," in G. Gigerenzer, L. Kruger, and M. Morgan, eds., *The Probabilistic Revolution: Ideas in Modern Science*, vol. 2 (Cambridge, Mass.: MIT Press, 1987), pp. 35–47.

35 B. T. Baldwin et al., *A Survey of Psychological Investigations with Reference to Differentiations between Psychological Experiments and Mental Tests* (Swarthmore, Penn.: APA Committee on the Academic Status of Psychology, 1916).

36 The problem is discussed at length in J. T. Lamiell, *The Psychology of Personality: An Epistemological Inquiry* (New York: Columbia University Press, 1987). See also K. Danziger, "The methodological imperative in psychology," *Philosophy of the Social Sciences* 15 (1985):1–13.

37 See H. Cravens, *The Triumph of Evolution: American Scientists and the Heredity–Environment Controversy 1900–1941* (Philadelphia: University of Pennsylvania Press, 1978), and S. J. Gould, *The Mismeasure of Man* (New York: Norton, 1981).

38 The use of treatment groups for purposes of experimental control was virtually unknown in experimental psychology prior to the explosion of educational research using such experimental design; see R. L. Solomon, "An extension of control group design," *Psychological Bulletin* 46 (1949):132–150. Even investigators like Thorndike were slow to appreciate the value of this approach. As Boring has pointed out, in their classical study of transfer of training, E. L. Thorndike and R. S. Woodworth, "The influence of improvement in one mental function upon

the efficiency of other functions,” *Psychological Review* 8 (1901):247–261, the authors used a control group only in a subsidiary experiment to which they devoted less than 1 percent of their article. Nor did the situation change during the next fifteen years. See E. G. Boring, “The nature and history of experimental control,” *American Journal of Psychology* 67 (1954):573–589.

39 Crude statistical comparison of groups subjected to different conditions or treatments was well known in nineteenth-century medicine, but there is no evidence that this had any influence on psychological practice. For the medical background, see A. M. Lilienfeld, “*Ceteris Paribus*: The evolution of the clinical trial,” *Bulletin of the History of Medicine* 56 (1982):1–18.

40 R. H. Winch, “The transfer of improvement of memory in schoolchildren,” *British Journal of Psychology* 2 (1908):284–293; Winch, “Some measurements of mental fatigue in adolescent pupils in evening schools,” *Journal of Educational Psychology* 1 (1910):13–24 and 83–100; Winch, “Accuracy in ‘transfer’?” *Journal of Educational Psychology* 1 (1910):557–589; Winch, “Mental fatigue in day school children, as measured by arithmetical reasoning,” *British Journal of Psychology* 4 (1911):315–341; Winch, “Mental fatigue in day school children as measured by immediate memory,” *Journal of Educational Psychology* 3 (1912):18–28 and 75–82; Winch, “Mental adaptation during the school day as measured by arithmetical reasoning,” *Journal of Educational Psychology* 4 (1913):17–28 and 71–84.

41 Winch seems to have been aware of his status as a pioneer. In a section entitled “The scientific interest of the work” he states: “I have, I believe for the first time, employed the method of *equal groups* to the solution of questions of fatigue” (Winch, *Journal of Educational Psychology* 1[1910]:16; emphasis in original). He was also given due credit for what soon came to be known as the “method of equivalent groups.”

42 J. P. Gilbert, “An experiment on methods of teaching zoology,” *Journal of Educational Psychology* 1 (1910):321–332; M. E. Lakeman, “The whole and part method of memorizing poetry and prose,” *Journal of Educational Psychology* 4 (1913):189–198; L. W. Kline, “Some experimental evidence in regard to formal discipline,” *Journal of Educational Psychology* 5 (1914):259–266; G. C. Myers, “Learning against time,” *Journal of Educational Psychology* 6 (1915):115–116; T. E. Mayman, “An experimental investigation of the book, lecture and experimental methods of teaching physics in elementary schools,” *Journal of Educational Psychology* (1915):6, 246–250. Peter Shermer (personal communication) has pointed out that the paper by Kline probably contains the first use of the term “control group” in the psychological literature.

43 E. L. Thorndike, “Fatigue in a complex function,” *Psychological Review* 21 (1914):402; H. E. Conrad and G. F. Arps, “An experimental study of economical learning,” *American Journal of Psychology* 27 (1916):507–529.

44 G. Melcher, “Suggestions for experimental work,” *Yearbook, National Society for the Study of Education*, 17, pt. 2 (1918):139–151. See also R. R. Rusk, *Experimental Education* (London and New York: Longmans, Green, 1919), pp. 10–11.

45 W. A. McCall, *How to Experiment in Education* (New York: Macmillan, 1923).

46 *Ibid.*, p. 3.

47 R. A. Fisher, *Statistical Methods for Research Workers* (London: Oliver & Boyd, 1925). McCall’s terminology was of course different from that of Fisher, but some of the techniques advocated were similar. The memory of McCall’s text was invoked by D. T. Campbell and J. C. Stanley in their enormously influential *Experimental*

and *Quasi-Experimental Designs for Research* (Chicago: Rand McNally, 1963). Randomization, another component of the new literature on experimental design, had a different background. See I. Hacking, "Telepathy: Origins of randomization in experimental design," *Isis* 79 (1988):427–451.

48 R. M. W. Travers, *How Research Has Changed American Schools* (Kalamazoo, Mich.: Mythos, 1983), pp. 154–157, 526.

49 For a detailed analysis of the paradigm case for experimenting in an industrial setting, see R. Gillespie, "The Hawthorne experiments and the politics of experimentation," in J. G. Morawski, ed., *The Rise of Experimentation in American Psychology* (New Haven, Conn.: Yale University Press, 1988), pp. 114–137.

50 Although laboratory investigators were slow to adopt treatment group methodology, the method became steadily more popular in the academic research literature. Use of treatment groups in the *Journal of Experimental Psychology* increased from only 6 percent of empirical studies in 1916–1920 to 15 percent in 1924–1926 and then to 23 percent in 1934–1936. The corresponding figures for the more conservative *American Journal of Psychology* were 2 percent (1914–1916), 8 percent (1924–1926), and 13 percent (1934–1936).

51 The idea of general laws of behavior that would be applied to social problems was a favorite theme of behaviorism. According to its founder, psychology should "establish laws or principles for the control of human action so that it can aid organized society in its endeavor to prevent failures in such adjustments." J. B. Watson, "An attempted formulation of the scope of behavior psychology," *Psychological Review* 24 (1917):329–352. But this perspective was by no means limited to behaviorists: "And it is the outstanding feature of our reconstructed psychology that it realized and accepted the obligation to apply to education, to social relations, to practical affairs, to the control of human behavior generally, normal and abnormal, desirable and undesirable, the conclusions arising from the scientific study of the mental side of man." See J. Jastrow, "The reconstruction of psychology," *Psychological Review* 34 (1927):169–195 (p. 170).

52 An occasional voice pointed out that the emperor's new statistical clothes could not hide his psychological nakedness: "A large part of the quantitative and statistical methodology now to be found in practical fields is associated rather with its use by certain psychologists and among biometricians than derived through a logical connection with psychology itself." Madison Bentley in C. Murchison, ed., *A History of Psychology in Autobiography*, vol. 3 (Worcester, Mass.: Clark University Press, 1936) p. 64. However, it is clear that Bentley was, and perceived himself as being, out of step with most of his fellow psychologists. The general issue is taken up again in chapter 9 of the present volume.

53 Part of the strong attraction that statistical theories of experimental design held for a later generation of experimenters was probably due to the possibilities they offered for transforming problems of individual history into statistical problems.

Chapter 8. Investigative practice as a professional project

1 The term "professional project" functions as a convenient shorthand for a complex of common interests and strategies for pursuing them. An extensive discussion of this complex is provided in M. S. Larson, *The Rise of Professionalism: A Sociological Analysis* (Berkeley: University of California Press, 1977). On the emergence of a professional identity among American psychologists, see T. M. Camfield, "The professionalization of American psychology," *Journal of the History of the Behavioral Sciences* 9 (1973):66–75.

2 Such concerns were far from being unique to psychologists. See R. Whitley, *The Intellectual and Social Organization of the Sciences* (Oxford: Clarendon Press, 1984).

3 On the “rhetorical fabric” that enveloped modern American psychology from its earliest years, see D. E. Leary, “Telling likely stories: The rhetoric of the new psychology, 1880–1920,” *Journal of the History of the Behavioral Sciences* 23 (1987):315–331.

4 It needs emphasizing that the essential conflict was systemic and not a conflict of persons. Individuals might therefore express a mixture of interests, depending on specific circumstances. The resulting tensions could be displayed on an intra-personal as well as an interpersonal level. But they were an inherent feature of the psychological enterprise in this particular historical context. Relevant material is to be found in J. M. O'Donnell, “The crisis of experimentalism in the 1920s,” *American Psychologist* 34 (1979):289–295; M. Sokal, “James McKeen Cattell and American Psychology in the 1920s,” in J. Brozek, ed., *Explorations in the History of American Psychology* (Lewisburg, Penn.: Bucknell University Press, 1982), pp. 273–323; and D. S. Napoli, *Architects of Adjustment: The History of the Psychological Profession in the United States* (Port Washington, N.Y.: Kennikat Press, 1981).

5 The organizational ups and downs of this relationship have been described in D. S. Napoli, *Architects of Adjustment*; also in J. E. W. Wallin, “History of the struggles within the American Psychological Association to attain membership requirements, test standardization, certification of psychological practitioners, and professionalization,” *Journal of General Psychology* 63 (1960):287–308; and partly in S. W. Fernberger, “The American Psychological Association: A historical summary, 1892–1930,” *Psychological Bulletin* 29 (1932):1–89. See also E. R. Hilgard, *Psychology in America: A Historical Survey* (San Diego: Harcourt Brace Jovanovich, 1987) chap. 20.

6 The notion that technological change was due to the application of science was part of the popular rhetoric of science at the time, and psychologists were able to deploy it effectively because it was such a pervasive illusion. See M. H. Rose, “Science as an idiom in the domain of technology,” *Science and Technology Studies* 5 (1987):3–11; and also the relevant parts of D. Bakan, “Politics and American psychology,” in R. W. Rieber and K. Salzinger, eds., *Psychology: Theoretical–Historical Perspectives* (New York: Academic Press, 1980), pp. 125–144.

7 “Every psychologist knows how difficult it is to get appropriations and maintenance for purely scientific work, and how much more impressive to the powers that control money is something which is ‘practical,’ however flimsy and evanescent its ‘practicality.’ The amount of money wasted in practical work which might be saved if more were available for the fundamental work on which eventual practical applications depend, is, of course, enormous, and even in psychology it is relatively large.” See K. Dunlap, “The experimental methods of psychology,” in M. Bentley et al., *Psychologies of 1925* (Worcester, Mass.: Clark University, 1926) p. 342. Note how the author deploys the model of technology as applied science to defend the priority of basic research on purely economic grounds.

8 On the role of experimentalism as an emblem of high status within the hierarchy of the sciences, see Whitley, *Intellectual and Social Organization*, pp. 276–289. One institutional mechanism through which such a hierarchy of prestige maintained itself was to be found in self-selected national scientific bodies. During the period of primary interest here, only one (token?) nonexperimentalist (Terman) was elected to the National Academy of Sciences, as compared with seven other psy-

chologists who were able to rely on their achievements in the laboratory (Angell, Dodge, Pillsbury, Seashore, Stratton, Woodworth, Yerkes).

9 It should be remembered that the coding priorities will tend to lead to an underestimation of the overall use of natural groups because only those studies that rely solely on this type of data were coded as such.

10 Whitley, *Intellectual and Social Organization*.

11 Sokal, "James McKeen Cattell."

12 In the public bureaucracy of education, with its traditional respect for academic qualifications, the educational psychologists were rather better protected from this threat. They were the designers and evaluators of mental tests, not their administrators. That task was relegated to a low-status stratum of "school psychologists" and teachers whose freedom of action was seriously circumscribed by the bureaucratic context in which they had to operate.

13 In any case there was a fundamental convergence between the kind of social situation in which the now dominant style of psychological experimentation was carried out and the social situation in mental testing. In both cases the situation was marked by an asymmetrical relationship in which the investigator was in a position of control and produced the kind of knowledge that was relevant to those in controlling positions. See the discussion of social practices of investigation in chapter 4. The issue is taken up again in chapter 11.

14 K. Danziger, "Statistical method and research practice in American psychology," in G. Gigerenzer, L. Kruger, and M. Morgan, eds., *The Probabilistic Revolution: Ideas in Modern Science*, vol. 2 (Cambridge, Mass.: MIT Press, 1987), pp. 35–47.

15 J. M. O'Donnell, *The Origins of Behaviorism: American Psychology, 1870–1920* (New York: New York University Press, 1985).

16 Perhaps the most insightful of these critics was Madison Bentley who expressed himself as follows in his mid–1930s autobiographical statement: "These new changes are in part the outgrowths of two decades when pressure to extend psychology toward the practical arts came chiefly from those arts themselves and not from a growing abundance of facts and principles within psychology itself." He goes on to criticize the common practice "of merely translating situations which arise in business, advertising, mental disorder, crime, delinquency, and so on, into terms which possess the flavour instead of the essence of a well-integrated science." See M. Bentley, in Carl Murchison, ed., *A History of Psychology in Autobiography*, vol. 3 (Worcester, Mass.: Clark University Press, 1936), pp. 63–64. A few years before, C. H. Judd had complained in a similar retrospective statement that "psychology is paying the price of its popularity." He counseled that "it will have to be cured of the idea that a hundred loosely related facts can be welded into a body of scientific truth by averaging discordant tendencies and covering up most of the facts by statistical juggling." See C. H. Judd, in Carl Murchison, ed., *A History of Psychology in Autobiography*, vol. 2 (Worcester, Mass.: Clark University Press, 1932), p. 230.

17 O'Donnell, "Crisis of Experimentalism." Boring's role as a critic of the misuse of statistics in psychology is discussed in chapter 9 of the present volume.

18 R. C. Tobey, *The American Ideology of National Science, 1919–1930* (Pittsburgh: University of Pittsburgh Press, 1971).

19 In F. Samelson, "Organizing for the kingdom of behavior: Academic battles and organizational policies in the twenties," *Journal of the History of the Behavioral Sciences* 21 (1985):33–47. The Carnegie Foundation had been financing mental

testing on a very large scale and also supported the research of key representatives of the new style of psychological research, like Terman and Thorndike, in a very generous way; see M. Russel, "Testers, trackers and trustees: The ideology of the intelligence testing movement in America 1900–1954" (Ph.D. diss., University of Illinois at Urbana–Champaign, 1972), chap. 3.

20 For an analysis of a particularly important instance of the process by which a formalistic kind of theory emerged "in the shadow of functionalist and pragmatic rhetoric," see J. G. Morawski, "Organizing knowledge and behavior at Yale's Institute of Human Relations," *Isis* 77 (1986):219–242.

21 On the role of industrial psychology in Germany, see S. Jaeger and I. Staeuble, "Die Psychotechnik und ihre gesellschaftlichen Entwicklungsbedingungen," in F. Stoll, ed., *Die Psychologie des 20. Jahrhunderts*, vol. 13 (Zurich: Kindler, 1980), pp. 53–95. Also S. Jaeger, "Zur Herausbildung von Praxisfeldern der Psychologie bis 1933," in M. G. Ash and U. Geuter, eds., *Geschichte der deutschen Psychologie im 20. Jahrhundert* (Opladen: Westdeutscher Verlag, 1985), pp. 83–112.

22 W. Stern, *Die differentielle Psychologie in ihren methodischen Grundlagen* (Leipzig: Barth, 1911); W. Stern, *Die psychologischen Methoden der Intelligenzprüfung und deren Anwendung an Schulkindern* (Leipzig: Barth, 1912); W. Betz, "Ueber Korrelation: Methoden der Korrelationsberechnung und kritischer Bericht über Korrelationsuntersuchungen aus dem Gebiete der Intelligenz, der Anlagen und ihrer Beeinflussung durch äussere Umstände," *Beihefte zur Zeitschrift für angewandte Psychologie und psychologische Sammelforschung* 3 (1911):1–80; G. Deuchler, "Ueber die Methoden der Korrelationsrechnung in der Pädagogik und Psychologie," *Zeitschrift für pädagogische Psychologie und experimentelle Pädagogik* 15 (1914):114–131, 145–159, 229–242. Spearman studied at Leipzig and collaborated with Felix Krueger, who was to become Wundt's successor; see F. Krueger and C. Spearman, "Korrelation zu verschiedenen geistigen Leistungsfähigkeiten," *Zeitschrift für Psychologie* 44 (1906):50–114.

23 P. Lundgreen, *Sozialgeschichte der deutschen Schule im Ueberblick*, Teil 1 (Göttingen: Vandenhoeck & Ruprecht, 1980).

24 P. Lundgreen, *ibid.*, Teil 2 (Göttingen: Vandenhoeck & Ruprecht, 1981).

25 For the specific considerations that underlie this highly synoptic statement, see F. K. Ringer, *Education and Society in Modern Europe* (Bloomington: Indiana University Press, 1979), chap. 5.

26 Mental tests, the main product on this market, were sometimes regarded with suspicion by classroom teachers because, although they sometimes allowed teachers to rationalize their failures, they also diminished the authority of their judgments of ability and because administrators used test results to measure the performance of teachers. See P. D. Chapman, "Schools as Sorters: Lewis M. Terman and the Intelligence Testing Movement, 1890–1930" (Ph.D. diss., Stanford University, 1979), pp. 185ff. This source also provides a cross-country comparison of test development (p. 167) which shows the early and overwhelming predominance of the United States in this field.

27 On the foundation of the Leipzig research institute and its significance for educational research in Germany, see the following: K. Ingenkamp, "Das Institut des Leipziger Lehrervereins 1906–1933 und seine Bedeutung für die Empirische Pädagogik," *Empirische Pädagogik* 1 (1987):60–70; C. C. Kohl, "The Institute for Experimental Pedagogy and Psychology of the Leipzig Lehrerverein," *Journal of Educational Psychology* 4 (1913):367–369; also P. Schlager, "Das Institut für experimentelle Pädagogik und Psychologie," *Pädagogisch-Psychologische Arbeiten* 1

(1910):V–XII; and a report in *Zeitschrift für experimentelle Pädagogik* 7 (1908):218–223. Vol. 10 (1910):133–139 of the same journal contains a report on the Berlin Institute. Subsequent volumes of *Pädagogisch-Psychologische Arbeiten* and of the *Zeitschrift für pädagogische Psychologie und experimentelle Pädagogik* contain regular reports on the activities of these and other similar institutes, including their budgets and lists of their supporters.

28 The Leipzig teachers' association published its own research journal, *Pädagogisch-Psychologische Arbeiten*, from 1910. Of the twenty empirical studies published there between 1910 and 1916 only one relied predominantly on group data, while eleven relied mainly on individual data, the rest being of the mixed type. In the 1920s group data studies reach half of the total.

29 Prominent examples of this pattern are provided by Karl Marbe and William Stern. See their autobiographical statements in Carl Murchison, ed., *A History of Psychology in Autobiography*, vol. 1 (Worcester, Mass.: Clark University Press, 1930), pp. 335–388, and vol. 3 (Worcester, Mass.: Clark University Press, 1936), pp. 181–213. The ambiguity of Stern's position when faced with practical questions emerges rather clearly in W. Stern, "Der personale Faktor in Psychotechnik und praktischer Psychologie," *Zeitschrift für angewandte Psychologie* 44 (1933):52–63; for an overview in English there is his *General Psychology from the Personalistic Standpoint* (New York: Macmillan, 1938); I. Staeuble, "William Stern's research program of differential psychology: Why did psychotechnics outstrip psychognostics?" a paper read at the 15th annual meeting of the Cheiron Society, Toronto, 1983.

30 See M. G. Ash, "Academic politics in the history of science: Experimental psychology in Germany, 1879–1941," *Central European History* 13 (1981):255–286. On the contrast between American and German psychology in this respect, see K. Danziger, "The social origins of modern psychology," in A. R. Buss, ed., *Psychology in Social Context* (New York: Irvington, 1979), pp. 27–45.

31 The terms in which an influential figure like E. Spranger opposed standardized tests were characteristic: "for the individual is viewed here in the end as something measurable and graspable in numbers, not as a structural principle of the soul." Quoted in M. G. Ash, "Psychology in twentieth-century Germany: Science and profession," in G. Cocks and K. Jarausch, eds., *German Professions 1800–1950* (New York: Oxford University Press, 1990).

32 The situation in psychology had its parallels in other disciplines. See J. Harwood, "National styles in science: Genetics in Germany and the United States between the World Wars," *Isis* 78 (1987):390–414.

33 D. Ruschemeyer, "Professionalisierung: Theoretische Probleme für die vergleichende Geschichtsforschung," *Geschichte und Gesellschaft* 6 (1980):311–325.

Chapter 9. From quantification to methodolatry

1 Cf. chapter 5, n. 14.

2 G. T. Fechner, *Elemente der Psychophysik* (Leipzig: Breitkopf & Härtel, 1860) and *In Sachen der Psychophysik* (Leipzig: Breitkopf & Härtel, 1877).

3 There is a comprehensive discussion of the issue, as it presented itself to the classical experimentalists in psychology, in E. B. Titchener, *Experimental Psychology: A Manual of Laboratory Practice* (New York: Macmillan, 1905), vol. 2, pt. 2. See also S. M. Stigler, *The History of Statistics: The Measurement of Uncertainty before 1900* (Cambridge, Mass.: Harvard University Press), pp. 253–254.

4 On Fechner's commitment to physical and mental atomism, see M. E. Marshall, "Physics, metaphysics, and Fechner's psychophysics," in W. R. Woodward and M. G. Ash, eds., *The Problematic Science: Psychology in Nineteenth-Century Thought* (New York: Praeger, 1982), pp. 65–87.

5 Much of the early literature on these so-called *Vexirfehler*, or paradoxical judgments, is reviewed in E. G. Boring, *Sensation and Perception in the History of Experimental Psychology* (New York: Appleton-Century-Crofts, 1942), pp. 479–480.

6 For a perceptive account of Binet's critical contributions to this issue, see T. H. Wolf, *Alfred Binet* (Chicago: University of Chicago Press, 1973), pp. 110–115.

7 See the series of studies on the two-point threshold published by Alfred Binet in volume 9 of *L'Année psychologique*. A translated excerpt from these has been published in R. H. Pollack and M. W. Brenner, eds., *The Experimental Psychology of Alfred Binet: Selected Papers* (New York: Springer, 1969), pp. 13–27. Prior to this the problems had been investigated quite systematically by Tawney, although not with Binet's fundamental critical insight; G. A. Tawney, "Ueber die Wahrnehmung zweier Punkte mittelst des Tastsinnes, mit Rücksicht auf die Frage der Uebung und die Entstehung der Vexirfehler," *Philosophische Studien* 13 (1897):163–221, and "Ueber die Trugwahrnehmung zweier Punkte bei der Berührung eines Punktes der Haut," *Philosophische Studien* 11 (1895):394–405.

8 This useful term was introduced by Nicholas Georgescu-Roegen in *The Entropy Law and the Economic Process* (Cambridge, Mass.: Harvard University Press, 1971), chap. 2, to refer to discrete and nonoverlapping concepts. I use the term only to refer to such concepts when they are also serially arranged.

9 This particular series is due to E. J. Gates, "The determination of the limens of single and dual impression by the method of constant stimuli," *American Journal of Psychology* 26 (1915):152–157; references to other, more complicated variants will be found in Boring, *Sensation and Perception*, p. 481.

10 E. B. Titchener, "On ethnological tests of sensation and perception with special reference to the tests of color vision and tactile discrimination described in the reports of the Cambridge Anthropological Expedition to Torres Straits," *Proceedings of the American Philosophical Society* 55 (1916):204–236.

11 Boring, *Sensation and Perception*, p. 482.

12 The best-known examples were those offered by William James: "To introspection, our feeling of pink is surely not a portion of our feeling of scarlet; nor does the light of an electric arc seem to contain that of a tallow-candle in itself" (*Principles of Psychology*, vol. 1 [New York: Henry Holt, 1890], p. 546). In his criticism of psychophysics published in 1887, the widely read philosopher, Henri Bergson, had generalized thus: "The fact is that there is no point of contact between the unextended and the extended, between quality and quantity. We can interpret the one by the other, set up the one as equivalent of the other; but sooner or later, at the beginning or at the end, we shall have to recognize the conventional character of the assimilation." See *Essai sur les données immédiates de la conscience*, translated as *Time and Free Will* in 1910 (New York: Harper Torchbook edition, 1960), p. 70. Also, W. W. Meissner, "The problem of psychophysics: Bergson's critique," *Journal of General Psychology* 66 (1962):301–309. Natural scientists also had their doubts; see J. von Kries, "Ueber die Messung intensiver Grössen und über das sogenannte psychophysische Gesetz," *Vierteljahresschrift für wissenschaftliche Philosophie und Soziologie* 6 (1882):257–294.

13 The classical review of this literature is E. G. Boring, "The stimulus error,"

American Journal of Psychology 32 (1921):449–471, reprinted in E. G. Boring, *History, Psychology, and Science: Selected Papers* (New York: Wiley, 1963). For a modern reconsideration, see G. A. Hornstein, “Quantifying psychological phenomena: Debates, dilemmas and implications,” in J. G. Morawski, ed., *The Rise of Experimentation in American Psychology* (New Haven, Conn.: Yale University Press, 1988), pp. 1–34. When psychologists began to measure *attitudes* as well as sensations, the classical objections resurfaced, though without any practical effect; see H. M. Johnson, “Pseudo-mathematics in the mental and social sciences,” *American Journal of Psychology* 48 (1936):342–351.

14 On the general problem of “measurement by fiat” in the social sciences, see A. V. Cicourel, *Method and Measurement in Sociology* (Glencoe, Ill.: The Free Press, 1964), chap. 1.

15 H. Ebbinghaus, *Memory: A Contribution to Experimental Psychology*, trans. H. A. Ruger and C. E. Bussenius (New York: Teachers College, Columbia University, 1913), from the German original, 1885.

16 That there was common ground between Wundt and Ebbinghaus emerges quite clearly when one compares their methods with other early attempts at an empirical study of memory: e.g., F. Galton, “Psychometric experiments,” *Brain* 2 (1879):149–162; J. Delboeuf, “La mémoire chez les hypnotisées,” *Revue philosophique* 21 (1886):441–472. The ideal of experimental control, both on the stimulus and on the response side, was much more intensively applied by the German investigators. This was not easily combined with an investigation of memory in its nonreproductive aspects.

17 In Wundt’s laboratory this function was represented as a logarithmic one with two “constants” whose value varied from individual to individual. K. H. Wolfe, “Untersuchungen über das Tongedächtnis,” *Philosophische Studien* 3 (1886):534–571.

18 W. Wundt, *Grundzüge der physiologischen Psychologie* (Leipzig: Engelmann, 1887), vol. 2, pp. 359–364.

19 K. Danziger, “Hermann Ebbinghaus and the psychological experiment,” in W. Traxel, ed., *Ebbinghaus Studien* 2 (Passau: Passavia, 1987), pp. 217–224.

20 S. Kvale, “Memory and dialectics: Some reflections on Ebbinghaus and Mao Tse-tung,” *Human Development* 18 (1975):205–222.

21 H. Ebbinghaus, *Urmanuskript “Ueber das Gedächtnis,”* with an introduction by W. Traxel (Passau: Passavia, 1983), p. 17. This is the original, unpublished version of Ebbinghaus’s monograph on his investigations of memory. On the importance of the serial form rather than the individual nonsense syllable for Ebbinghaus’s work, see also E. Scheerer, “Ebbinghaus, Herbart and Hegel; or: Not the syllable was meaningless, but the syllables series,” in W. Traxel, ed., *Ebbinghaus Studien* 2 (Passau: Passavia, 1987), pp. 9–22.

22 This is much more explicit in the draft version of his monograph on memory (see n. 21) than in the published version.

23 The metaphor of psychological activity as a kind of work was also very important for other nineteenth-century pioneers of modern psychology, especially Fechner, whose notion of psychophysics is inconceivable without it. The historical significance of the energy model is discussed extensively in S. Jaeger and I. Staeuble, *Die gesellschaftliche Genese der Psychologie* (Frankfurt: Campus, 1978), pp. 293–315.

24 E. Meumann, *Abriss der experimentellen Pädagogik* (Leipzig: Teubner, 1912).

25 Ibid., pp. 240–242.

26 Meumann contextualizes his research interests as follows: “In the traditional pedagogy we read a great deal about methods of teaching; but in most cases the pedagogical text-books can tell us nothing about methods of learning. And yet we find ourselves confronted by the very serious question as to whether the efficiency of school-room management may not be increased by systematically improving the pupil’s procedure in the act of learning in such a fashion that his learning may be perfected in its technical aspects and accomplished more economically. This question becomes the more pressing in modern times because our courses of study, in their attempt to comply with the increasing requirements of practical life, are becoming more and more exacting in the demands which they make upon the memory tasks of school children.” E. Meumann, *The Psychology of Learning: An Experimental Investigation of the Economy and Technique of Memory* (New York: Appleton, 1913), p. xiv. (This is an English translation of the third edition of a German work that had first appeared in 1907.)

27 Meumann, *Psychology of Learning*, pp. xvi–xvii. On the emergence of German “experimental pedagogics” and its relationship to Ebbinghaus’s research, see S. Sporer, “Gedächtnis in vitro und in vivo: von Hermann Ebbinghaus’ sinnlosen Silben zur experimentellen Pädagogik und zur Aussagepsychologie,” in W. Traxel, ed., *Ebbinghaus Studien 2* (Passau: Passavia, 1987), pp. 107–119.

28 Meumann, *Psychology of Learning*, pp. 232, 372–373. The problem was complicated in practice by the need to take into account the appearance of what was called mental or psychological fatigue. Too much fatigue detracted from the economy of learning. Despite numerous attempts in the early years of the century, the phenomena grouped under the heading of “fatigue” resisted the imposition of a satisfactory numerizing structure. Such attempts were renewed later by Clark Hull, whose neobehaviorist translation of fatigue into “inhibition” did not however lead to any greater success with the measurement problem.

29 There is considerable overlap in the content of Meumann’s *Psychology of Learning* and the second volume of Thorndike’s *Educational Psychology*, entitled *The Psychology of Learning* (New York: Teachers College, Columbia University, 1913). It is the first volume of Thorndike’s text, with its psychobiological orientation, that has no real counterpart in Meumann’s work. That orientation established a necessary link for Thorndike, and other American psychologists of his generation, between their day-to-day work and the most grandiose ambitions: “It will, of course, be understood that directly or indirectly, soon or late, every advance in the sciences of human nature will contribute to our success in controlling human nature and changing it to the advantage of the common weal,” writes Thorndike in an article on “The contributions of psychology to education,” in *The Journal of Educational Psychology* 1 (1910):5–12 (p. 8).

30 In the preface to the second edition of his *Vorlesungen zur Einführung in die experimentelle Pädagogik und ihre psychologischen Grundlagen* (Introductory Lectures on Experimental Pedagogics and Its Psychological Foundations) (Leipzig: Engelmann, 1911), Meumann lists ten teachers’ associations (*Lehrervereine*) for which he had prepared this material. On the teachers’ associations, see chapter 8 of the present volume.

31 Merle Curti in *The Social Ideas of American Educators* (Paterson, N.J.: Littlefield, Adams, 1961), p. 460, considers that Thorndike’s “influence in establishing and popularizing the fact-finding, statistical, and experimental technique in education has been immeasurable.”

32 The enormous extension in the application of the learning concept did not

however affect the interpretation of learning as a kind of work. The claim that all adaptive behavior was governed by universal “principles of learning” simply had the effect of giving the work model the status of a fundamental biological truth. A secularized Calvinism may have played a role here. “My God does not hear prayers. Work for him,” proclaims the character who speaks for Thorndike in a dramatic script he never published. See C. J. Karier, *Scientists of the Mind: Intellectual Founders of Modern Psychology* (Urbana and Chicago: University of Illinois Press, 1986), chap. 3.

33 The quoted statement occurs at the beginning of what Thorndike refers to as a “general *Credo*” in a chapter entitled “The nature, purposes, and general methods of measurements of educational products” that formed part of the *17th Yearbook of the National Society for the Study of Education*, pt. 2 (Bloomington, Ill.: Public School Publishing Co., 1918). Some statements commonly attributed to Thorndike on this issue were in fact simplified restatements by disciples. His own attitude is perfectly conveyed by the following statement: “History records no career, war or revolution that can compare in significance with the fact that the correlation between intellect and morality is approximately .3, a fact to which perhaps a fourth of the world’s progress is due.” See “Educational diagnosis,” *Science* 37 (1913):142.

34 See G. Joncich, *The Sane Positivist: A Biography of Edward L. Thorndike* (Middletown, Conn.: Wesleyan University Press, 1968).

35 According to Thorndike, “the facts and laws of psychology . . . should provide the general basis for the interpretation and explanation of the great events studied by history, the complex activities of civilized society, the motives that control the action of labor and capital, and the causes to which linguistic inventions and modifications are due.” See *The Elements of Psychology*, 2d ed. (New York: Seiler, 1907), p. 324.

36 Thorndike consistently advocated the abdication of individual judgment in the face of the authority of experts: “Outside that field [our own expertise] the intelligent procedure for most of us is to refuse to think, spending our energy rather in finding the expert in the case and learning from him.” From “The psychology of the half-educated man,” *Harper* 140 (April 1920):670.

37 Thorndike’s conservative identification of the prevailing social order with the order of nature was extreme, even for his generation; see Joncich, *The Sane Positivist*, and Karier, *Scientists of the Mind*. During the Great Depression he wrote: “When it is not feasible to learn what the consequences of weighting one person’s satisfactions more than another’s will be, our trustee for humanity will do well to weight the wants of good men more than the same wants of bad men, since there is a probability that the gratification of wants will cause both to maintain or increase their customary activities” (*Science* 83 [1936]:1–8).

38 W. G. Sleight, “Memory and formal training,” *British Journal of Psychology* 4 (1911):386–457. Twelve years earlier Binet’s associate, Victor Henri, had made a general suggestion about using an estimate of variability to establish the significance of the difference between group averages: V. Henri, “Quelques applications du calcul des probabilités à la psychologie,” *L’Année Psychologique* 5 (1899):153–160. However, at that point there was no group of psychological investigators for whose work the suggestion had practical relevance, and so it was ignored and remained buried in the literature. It is not even mentioned in a 1929 bibliography compiled for psychologists and educationists that contains many older and less directly relevant items; see H. Walker, *Studies in the History of Statistical Method* (Baltimore: Williams & Wilkins, 1929).

39 E.g., W. H. Winch, “Experimental researches on learning to spell,” *Journal of*

Educational Psychology 4 (1913):525–537 (p. 535). Other investigators also took notice of the statistical weakness of Winch's earlier studies and introduced the appropriate techniques into their own work – e.g., E. J. G. Bradford, "A note on the relation and aesthetic value of the perceptive types in color appreciation," *American Journal of Psychology* 24 (1913):545–554 (p. 548).

40 For an example of graphical methods of group comparison, see W. G. Chambers, "Individual differences in grammar grade children," *Journal of Educational Psychology* 1 (1910):61–75.

41 J. R. McGaughy, *The Fiscal Administration of City School Systems* (New York: Macmillan, 1924).

42 A comparison of two widely used texts of the time with their earlier counterparts indicates that the treatment of this topic had become more prominent and extensive, and that consideration of less mechanical methods for assessing the degree of overlap between two distributions had been dropped. See H. Sorenson, *Statistics for Students of Psychology and Education* (New York: McGraw-Hill, 1936); H. E. Garrett, *Statistics in Psychology and Education*, 2d ed. (New York and London: Longmans, Green, 1937) (1st, 1926); T. L. Kelley, *Statistical Method* (New York: Macmillan, 1924); E. L. Thorndike, *An Introduction to the Theory of Mental and Social Measurements*, 2d ed. (New York: Science Press, 1919).

43 This development probably prepared the ground for the relatively rapid wholesale adoption of Fisherian techniques of significance testing and experimental design in the subsequent period. See A. J. Rucci and R. D. Tweney, "Analysis of variance and the 'second discipline' of scientific psychology: A historical account," *Psychological Bulletin* 87 (1980):166–184. At the same time, the methodological preconceptions formed during the earlier period led to certain difficulties in appreciating the special features of the Fisherian approach. See A. D. Lovie, "The analysis of variance in experimental psychology: 1934–1945," *British Journal of Mathematical and Statistical Psychology* 32 (1979):151–178. The outcome was a curious hybrid of Fisherian and non-Fisherian approaches. See G. Gigerenzer and D. J. Murray, *Cognition as Intuitive Statistics* (Hillsdale, N.J.: Lawrence Erlbaum, 1987), chap. 1.

44 J. O'Donnell, "The crisis of experimentalism in the 1920s: E. G. Boring and his use of history," *American Psychologist* 34 (1979):289–295; and D. A. Stout, "Personal politics or policing a profession? A methodological debate between E. G. Boring and Carl Murchison" (unpublished manuscript). Boring was not the only experimentalist who inveighed against the misuse of statistics encouraged by the predominance of applied interests; see, e.g., K. Dunlap, "The experimental methods of psychology," in M. Bentley et al., *Psychologies of 1925* (Worcester, Mass.: Clark University, 1926).

45 E. G. Boring, "Mathematical vs. scientific significance," *Psychological Bulletin* 16 (1919):335–338; "The logic of the normal law of error in mental measurement," *American Journal of Psychology* 31 (1920):1–33; "Scientific induction and statistics," *American Journal of Psychology* 37 (1926):303–307.

46 Boring, "Significance," p. 338.

47 Boring, "Logic," p. 33.

48 T. L. Kelley, "The principles and techniques of mental measurement," *American Journal of Psychology* 34 (1923):408–432 (p. 419).

49 The most explicit of these conventions concerned "acceptable" levels of statistical significance, which appear to have been extensions of older conventions. See M. Cowles and C. Davis, "On the origin of the .05 level of statistical significance,"

American Psychologist 37 (1982):553–558. Such conventions defining “reasonable agreement” are not of course peculiar to psychology; see T. S. Kuhn, “The function of measurement in modern physical science,” *Isis* 52 (1961):162–176.

50 In later years some critical voices were raised against this and related practices. See D. E. Morrison and R. E. Henkel, eds., *The Significance Test Controversy* (Chicago: Chicago University Press, 1970).

51 On the significance of these practices see G. Gigerenzer, “Probabilistic thinking and the fight against subjectivity,” in L. Krüger, G. Gigerenzer, and M. S. Morgan, eds., *The Probabilistic Revolution*, vol. 2.: *Ideas in the Sciences* (Cambridge, Mass.: MIT Press, 1987); also Gigerenzer and Murray, *Intuitive Statistics*, chap. 1; and M. C. Acree, “Theories of Statistical Inference in Psychological Research: A Historico-Critical Study” (Ph.D. diss., Clark University, 1978).

52 B. D. Mackenzie, *Behaviourism and the Limits of Scientific Method* (London: Routledge and Kegan Paul, 1977).

53 D. Bakan, *On Method* (San Francisco: Jossey-Bass, 1967) chaps. 1–3. This practice has also been troubling to that minority of psychologists who have perceived the inconsistency between it and traditional scientific determinism; see M. Sidman, *Tactics of Scientific Research* (New York: Basic Books, 1960).

54 W. S. Hunter, “Correlation studies with the maze in rats and humans,” *Comparative Psychology Monographs* 1 (1922), no. 1, pt. 2; W. S. Hunter and R. Vance, “Further studies on the reliability of the maze with rats and humans,” *Journal of Comparative Psychology* 4 (1924):431–442; H. Carr, “The reliability of the maze experiment,” *Journal of Comparative Psychology* 6 (1926):85–93; W. S. Hunter, “Reply to Professor Carr on ‘The reliability of the maze experiment,’” *Journal of Comparative Psychology* 6 (1926):393–398; R. Tryon, “Effect of the unreliability of measurement on the difference between groups,” *Journal of Comparative Psychology* 6 (1926):449–453; E. C. Tolman and D. B. Nyswander, “The reliability and validity of maze measures for rats,” *Journal of Comparative Psychology* 7 (1927):425–460; R. C. Tryon, “Demonstration of the effect of unreliability of measurement on a difference between groups,” *Journal of Comparative Psychology* 8 (1928):1–22. The general implications of the discussion were not lost on psychologists working outside this field; see H. R. Crosland, “Certain points concerning the reliabilities of experiments in psychology,” *American Journal of Psychology* 40 (1928):331–337.

55 K. Dunlap, “The average animal,” *Journal of Comparative Psychology* 19 (1935):1–3.

56 D. L. Krantz, “Schools and systems: The mutual isolation of operant and non-operant psychology as a case study,” *Journal of the History of the Behavioral Sciences* 8 (1972):86–102 (p. 92).

57 On the problems of imposing statistical models involving the continuity assumption on distributions that mask an underlying psychological discontinuity, see R. Harré, *Social Being* (Oxford: Blackwell, 1979), p. 108; also M. Oakes, *Statistical Inference: A Commentary for the Social and Behavioural Sciences* (Chichester and New York: Wiley, 1986), pp. 163–166. As pointed out in chapter 5, the issue goes back to the nineteenth-century roots of social statistics.

58 For further discussion of this issue, see K. Danziger, “The methodological imperative in psychology,” *Philosophy of the Social Sciences* 15 (1985):1–13.

59 P. E. Meehl, “Theory testing in psychology and physics: A methodological paradox,” *Philosophy of Science* 34 (1967):103–115; also Oakes, *Statistical Inference*, chap. 2.

60 To avoid possible misunderstanding I should add that this criticism is directed at the historically pervasive misuse of statistical inference, not at its use in connection with problems that are inherently statistical in nature.

61 K. Danziger, "On theory and method in psychology," in W. Baker, L. Mos, and H. Rappard, eds., *Recent Trends in Theoretical Psychology* (New York: Springer-Verlag, 1988), pp. 87–94.

62 See Gigerenzer and Murray, *Intuitive Statistics*, chaps. 2 and 5.

63 W. Wundt, *Outlines of Psychology* (Leipzig: Engelmann, 1897), sec. 24. A fuller statement appears in his *Logik*, 3d ed., vol. 3 (Stuttgart: Enke, 1908), pp. 274–282.

Chapter 10. Investigating persons

1 F. Samelson, "Putting psychology on the map: Ideology and intelligence testing," in A. R. Buss, ed. *Psychology in Social Context* (New York: Irvington, 1980), pp. 103–168.

2 R. T. von Mayrhauser, "The manager, the medic, and the mediator: The clash of professional psychological styles and the wartime origins of group mental testing," in M. M. Sokal, ed., *Psychological Testing and American Society* (New Brunswick, N.J.: Rutgers University Press, 1987), pp. 128–157.

3 J. R. McCrory, "A study of the relation between ability and achievement," *Educational Administration and Supervision* 12 (1926):481–490 (see p. 490).

4 See, e.g., A. T. Poffenberger and F. L. Carpenter, "Character traits in school success," *Journal of Experimental Psychology* 7 (1924):67–74; W. H. Hughes, "Organized personnel research and its bearing on high-school problems," *Journal of Educational Research* 10 (1924):386–398; G. E. Manson, "Personality differences in intelligence test performance," *Journal of Applied Psychology* 9 (1925):230–255; A. E. Traxler, "The Will–Temperament of upper-grade and high-school pupils," *School Review* 33 (1925):264–273. In 1921 (vol. 12), the *Journal of Educational Psychology* published a symposium on "Intelligence and its measurement." Many of the contributors called for an extension of mental testing to cover character and personality; see, e.g., Freeman, p. 139, Pintner, p. 142, Pressey, p. 147, Haggerty, p. 216, Buckingham, p. 275.

5 The history of intelligence testing had always been marked by a fundamental divergence between those who were guided by concrete practical goals, like Binet, and those with a commitment to the goals of eugenics, like most of the well-known American and British pioneers in the field. The latter tended to give some global significance to that which intelligence tests measured. In one sense the history of the field consists of a series of concessions forced upon the second group by the practical demonstrations of the first group. See, e.g., von Mayrhauser, "Manager, medic, and mediator"; T. M. Wolf, *Alfred Binet* (Chicago: University of Chicago Press, 1973); H. Cravens, *The Triumph of Evolution* (Philadelphia: University of Pennsylvania Press, 1978), chap. 7.

6 For a detailed consideration of these changes, see J. Parker, "From the Intellectual to the Non-intellectual Traits: A Historical Framework for the Development of American Personality Research" (M.A. thesis, York University, Toronto, 1986). I am much indebted to this source for the historical background of the issues discussed in this section.

7 Methodological implications of this practice are critically considered in J. T. Lamiell, *The Psychology of Personality* (New York: Columbia University Press,

1987). See also the classical analysis of the topic in P. Lafitte, *The Person in Psychology: Reality or Abstraction* (London: Routledge and Kegan Paul, 1957).

8 For a comprehensive critical discussion of this issue, see G. Gigerenzer, *Messung und Modellbildung in der Psychologie* (Munich: Reinhardt, 1981), pp. 88–112.

9 As an illustration, the personality category of extraversion–introversion certainly did not owe its popularity to the fact that many American psychologists held Jung's theoretical system in high esteem or were even especially interested in it. Rather, these terms were prised loose from their original theoretical integument and generally assimilated to prevailing popular conceptions of the healthy “well-adjusted” personality and its opposite.

10 J. Parker, “The evolution of trait concepts in American personality research: Trait terms used in the *Journal of Abnormal and Social Psychology*, 1920 to 1955,” a paper read at the 6th annual European Cheiron conference, Brighton, 1987.

11 S. W. Fernberger, “The scientific interests and scientific publications of the members of the American Psychological Association,” *Psychological Bulletin* 35 (1938):261–281. The year 1937 was marked by the appearance of G. W. Allport's *Personality: A Psychological Interpretation* (New York: Holt, Rinehart & Winston, 1937), an influential text that established the scope and definition of the new field for many years to come.

12 G. W. Allport, “Personality and character,” *Psychological Bulletin* 18 (1921):441–455; P. M. Symonds, “The present state of character measurement,” *Journal of Educational Research* 15 (1924):484–498.

13 For a comprehensive account, see C. M. Shea, “The Ideology of Mental Health and the Emergence of the Therapeutic Liberal State: The American Mental Hygiene Movement, 1900–1930” (Ph.D. diss., University of Illinois at Urbana-Champaign, 1980).

14 During the 1920s mental hygiene appears to have received about as much money from private foundations as the entire field of psychology. See table 11 in R. E. Kohler, “Science, foundations and American universities in the 1920's,” *Osiris* n.s. 3 (1987):135–164.

15 In this connection John Burnham speaks of “the concept of improving the world not by social action but by a patient program of individual treatment of large numbers of people.” See J. Burnham, “The new psychology: From narcissism to social control,” in J. Braeman, R. H. Brenner, D. Brody, eds., *Change and Continuity in Twentieth Century America: The 1920's* (Columbus: Ohio State University Press, 1968), pp. 351–398. Also Shea, “Ideology of Mental Health.”

16 The Laura Spelman Rockefeller Memorial Fund, established in 1918 with an endowment of \$74 million, played a crucial role here. See the archival material referred to in A. B. Smuts, “The National Research Council Committee on Child Development and the founding of the Society for Research in Child Development, 1925–1933,” in A. B. Smuts and J. W. Hagen, “History and Research in Child Development,” *Monographs of the Society for Research in Child Development* 50 (1985):4–5 (serial no. 211).

17 This reference of the term “personality” is to be clearly distinguished from several other meanings the term may take on in everyday discourse. Of all the slippery terms that define modern psychological discourse this one is perhaps the most slippery. On some historical variants, see R. Williams, *Keywords: A Vocabulary of Culture and Society* (London: Fontana, 1976); and W. I. Susman, “Personality and the making of twentieth-century culture,” in J. Higham and P. K.

Conkin, eds., *New Directions in American Intellectual History* (Baltimore: Johns Hopkins University Press, 1979), pp. 212–226.

18 In an analysis of the profound influence of the mental hygiene movement on American education Sol Cohen expresses this as follows: “The idea of the school’s responsibility for ‘personality development’ is a kind of ideogram which serves as shorthand for a systematically related cluster of attitudes, assumptions and practices with profound implications for curriculum, grading and promotion practices, methods of instruction, notions of achievement, authority, and discipline, and which also includes a tacit assumption by the schools of a broad parent–surrogate function.” S. Cohen, “The school and personality development: Intellectual history,” in J. H. Best, ed., *Historical Inquiry in Education: A Research Agenda* (Washington, D.C.: American Educational Research Association, 1983), p. 110. Also see S. Cohen, “The mental hygiene movement, the development of personality and the school: The medicalization of American education,” *History of Education Quarterly* 23 (1983):123–149.

19 P. Shermer, “The development of research practice in abnormal and social psychology: 1906–1956” (M.A. thesis, York University, Toronto, 1983). See also R. W. Lissitz, “A longitudinal study of the research methodology in the *Journal of Abnormal and Social Psychology*, the *Journal of Nervous and Mental Disease*, and the *American Journal of Psychiatry*,” *Journal of the History of the Behavioral Sciences* 5 (1969):248–255. The *Journal of Personality* (at first *Character and Personality*), which began to appear in the 1930s, showed a preference for statistical studies almost immediately. For more recent trends, see K. H. Craik, “Personality research methods: An historical perspective,” in *Journal of Personality* 54 (1986):18–51.

20 See L. Cochran, “On the categorization of traits,” *Journal for the Theory of Social Behaviour* 14 (1984):183–209.

21 P. E. Vernon, “The American v. the German methods of approach to the study of temperament and personality,” *British Journal of Psychology* 24 (1933):156–177.

22 W. Dilthey, “Ideen über eine beschreibende und zergliedernde Psychologie,” *Sitzungsberichte der Akademie der Wissenschaften zu Berlin* 2 (1894):1309–1407. (Reprinted in W. Dilthey, *Gesammelte Schriften*, vol. 5 [Stuttgart: Teubner, 1957], pp. 139–240.)

23 T. Lipps, *Psychologische Untersuchungen I* (Leipzig: Engelmann, 1907).

24 A few years previously Wundt had presented a somewhat similar theory; see W. Wundt, *Völkerpsychologie*, vol. 1 (Leipzig: Engelmann, 1900), and K. Danziger, “Origins and basic principles of Wundt’s *Völkerpsychologie*,” *British Journal of Social Psychology* 22 (1983):303–313.

25 Lipps’s views were summarized in his famous *empathy* concept (*Einfühlung*), a term that experienced some strange transmutations of meaning after its adoption by American psychologists. See K. Holzkamp, “Zur Geschichte und Systematik der Ausdruckstheorien,” in R. Kirchhoff, ed., *Ausdruckspsychologie* (Göttingen: C. J. Hogrefe, 1965), pp. 39–113; also G. A. Gladstein, “The historical roots of contemporary empathy research,” *Journal of the History of the Behavioral Sciences* 20 (1984):38–59.

26 L. Klages, *Die Grundlegung der Wissenschaft vom Ausdruck* (1907; repr. Bonn: Bouvier, 1964); and L. Klages, *Handschrift und Charakter* (1917; repr. Bonn: Bouvier 1968). The latter book enjoyed an extraordinary popularity for half a century, the 1968 edition being its twenty-sixth.

27 Graphology as a systematic method had emerged at the same time as experimental psychology. The term is said to have made its first appearance in J. H. Michon, *Système de Graphologie*, (Paris: Marpon et Flammarion, 1875).

28 See R. Kichhoff, "Zur Geschichte des Ausdrucksbegriffs" (1965), in Kirchhoff, *Ausdruckspsychologie*; also K. Bühler, *Ausdrucks-theorie* (Jena: G. Fischer, 1933).

29 This type of study is pursued in contemporary social psychology under the heading of "nonverbal communication."

30 The idea of "character" as a knowledge object to be accessed in a systematic and methodical way originates with J. Bahnsen, *Beiträge zur Characterologie mit besonderer Berücksichtigung pädagogischer Fragen* (Leipzig: Brockhaus, 1867). The practical context for this development was provided by the field of education.

31 See M. G. Ash "Academic politics in the history of science: Experimental psychology in Germany, 1879–1941," *Central European History* 13(1981):255–286.

32 G. Lukacz, *The Destruction of Reason*, trans. P. Palmer (London: Merlin, 1980), from the German original, 1954.

33 See S. Jaeger and I. Staeuble, "Die Psychotechnik und ihre gesellschaftlichen Entwicklungsbedingungen," in F. Stoll, ed., *Die Psychologie des 20. Jahrhunderts*, vol. 13 (Zurich: Kindler, 1980), pp. 53–95.

34 By 1929 there were eight candidates for every vacancy. After that the discrepancy increased. See L. von Renthe-Fink, "Von der Heerespsychotechnik zur Wehrmachtpsychologie" (1985) in *Deutsche Wehrmachtpsychologie 1914–1945* (München: Verlag für Wehrwissenschaften, 1985), p. 87. On the history of military psychology in Germany, see U. Geuter, "Polemos panton pater – Militär und Psychologie im Deutschen Reich 1914–1945," in M. G. Ash and U. Geuter, eds., *Geschichte der deutschen Psychologie im 20. Jahrhundert* (Opladen: Westdeutscher Verlag, 1985), pp. 146–171.

35 In actual fact, the introduction of psychological selection methods was correlated with a distinct *increase* in the proportion of successful candidates originating in the traditional strata from which officers had always been recruited. See D. Bald, "Sozialgeschichte der Rekrutierung des deutschen Offizierkorps von der Reichsgründung bis zur Gegenwart," *Schriftenreihe Innere Führung* (Reihe Ausbildung und Bildung) 29 (München: Sozialwissenschaftliches Institut der Bundeswehr, 1977).

36 For some details see H. L. Ansbacher, "German military psychology," *Psychological Bulletin* 38 (1941):370–392 (p. 383). This paper contains an excellent bibliography, although its own summary gives too much prominence to some aspects given little weight in the originals.

37 Von Renthe-Fink, "Heerespsychotechnik," p. 62.

38 The most influential of the systems of personality psychology developed by German army psychologists was that presented in P. Lersch, *Aufbau des Charakters* (Leipzig: Barth, 1938). His special practical field was the diagnostic use of facial expression. See his *Gesicht und Seele: Grundlinien einer mimischen Diagnostik* (1931; repr. München: Reinhardt, 1966). Other prominent academic psychologists for whose work the military experience was crucial were J. Rudert and K. Mierke.

39 See von Renthe-Fink, "Heerespsychotechnik."

40 This is a major conclusion of the very thorough study of the relationship between psychology and National Socialism presented in U. Geuter, *Die Professionalisierung der deutschen Psychologie im Nationalsozialismus* (Frankfurt: Suhrkamp, 1984).

41 See *ibid.*, pp. 180–186.

42 W. Fritscher, “Die psychologische Auswahl des Offiziersnachwuchses in der deutschen Wehrmacht” (1985), in *Deutsche Wehrmachtpsychologie*, pp. 421–475ff.

43 When the selection of an elite became part of the task of Allied psychologists during World War II, they adopted some key features of the German methods. See OSS Assessment Staff, *Assessment of Men* (New York: Rinehart, 1948).

44 Psychological procedures for the selection of officers were abolished by the German army in 1942, largely as a result of enormous losses on the Eastern Front and a drying up of the manpower reservoir. It now seemed more rational to rely on battle testing, rather than on psychological testing, to pick men who would do well under combat conditions.

45 See Geuter, *Die Professionalisierung*, pp. 202–205.

46 There were of course other types of innovative investigative practice in neighboring areas that did not form part of the discipline of psychology. Psychoanalysis and the methods of social research practiced by some members of the Frankfurt school are notable instances. On the latter, see E. Fromm, *The Working Class in Weimar Germany: A Psychological and Sociological Study* (Cambridge, Mass.: Harvard University Press, 1984).

47 Participating as an experimental subject in the investigations of other members of the group was part of the normal experience of those who belonged to this circle. See, e.g., Anita Karsten’s autobiographical statement in L. J. Pongratz, W. Traxel, and E. G. Wehner, eds., *Psychologie in Selbstdarstellungen*, vol. 2 (Bern: Huber, 1977), pp. 77–108.

48 T. Dembo, “Der Aerger als dynamisches Problem,” *Psychologische Forschung* 15 (1931):1–144 (pp. 73–84).

49 The distinction that is sometimes made between an earlier phase of Lewin’s research, directed at personality, and a later phase, directed at social psychology, can be misleading. Most of the “later” social psychological concepts, such as “power fields,” occur in the earlier work, as a glance at the research papers published by Lewin’s students will show.

50 K. Lewin, “Die Erziehung der Versuchsperson zur richtigen Selbstbeobachtung und die Kontrolle psychologischer Beschreibungsangaben,” in C. F. Graumann and A. Métraux, eds., *Kurt Lewin Werkausgabe* (Bern: Huber, and Stuttgart: Klett, 1981), pp. 153–212. This is a paper that Lewin wrote around 1918 but never published in his lifetime.

51 E.g., B. Zeigarnik, “Das Behalten erledigter und unerledigter Handlungen,” *Psychologische Forschung* 9 (1927):1–85 (pp. 18–19).

52 K. Lewin, “Vorbemerkungen über die psychischen Kräfte und Energien und über die Struktur der Seele,” *Psychologische Forschung* 7 (1926):294–329 (p. 304). It should be noted that the word *Versuch*, which occurs several times in this passage, could mean either an experiment or a trial. I have been guided by the general context of the article in translating this passage.

53 In his later writings Lewin elaborates this point into a full-blown “field theory”; see, e.g., his “Behavior and development as a function of the total situation” in K. Lewin, *Field Theory in Social Science* (London: Tavistock, 1952), pp. 238–303.

54 K. Lewin, “Die psychischen Kräfte,” pp. 306–307.

55 N. Ach, *Ueber den Willensakt und das Temperament* (Leipzig: Quelle & Meyer, 1910). For an extensive review of the introspective experimental study of volition, see J. Lindworsky, *Der Wille: Seine Erscheinung und seine Beherrschung nach den*

Ergebnissen der experimentellen Forschung (Leipzig: Barth, 1923). A modern introduction is provided by H. Gundlach, "Anfänge der experimentellen Willenspsychologie," in H. Heckhausen, ed., *Jenseits des Rubikon: Der Wille in den Humanwissenschaften* (Berlin, Heidelberg: Springer-Verlag, 1987), pp. 67–85.

56 K. Lewin, *Die Entwicklung der experimentellen Willenspsychologie und die Psychotherapie*, (Leipzig: Hirzel, 1929), p. 12.

57 In view of Lewin's explicit rejection of phenomenology, a description of his Berlin research program as "experimental phenomenology" seems rather inappropriate. Apart from this unfortunate choice of terminology, the editor's comments in J. de Rivera, *Field Theory as Human-Science: Contributions of Lewin's Berlin Group* (New York: Gardner, 1976), provide a very useful English-language introduction to the topic.

58 Some of his work in this area was reflected in his dissertation, republished as vol. 2 of his collected works. See C. F. Graumann and A. Métraux, eds., *Kurt Lewin Werkausgabe*, vol. 2 (Bern: Huber, and Stuttgart: Klett, 1983).

59 On Lewin's distinction between genotype and phenotype, see "The conflict between Aristotelian and Galilean modes of thought in contemporary psychology," in K. Lewin, *Dynamic Theory of Personality* (New York: McGraw-Hill, 1935), p. 11.

60 Lewin speaks of "the old, mistaken theory of induction" in discussing the nature of psychological experiments. See K. Lewin, "Gesetz und Experiment in der Psychologie," *Symposion 1* (1927):375–421 (p. 384), republished in Graumann and Métraux, *Kurt Lewin Werkausgabe*, vol. 1, pp. 279–320.

61 K. Lewin, "Gesetz und Experiment," p. 291; Lewin discusses the same point in slightly different terms in "Aristotelian and Galilean modes of thought."

62 Lewin, "Aristotelian and Galilean modes of thought."

63 K. Lewin, "Gesetz und Experiment," p. 406.

64 By far the best English-language source of information on these studies is to be found in de Rivera, *Field Theory as Human-Science*

65 See D. Cartwright, "Theory and practice," *Journal of Social Issues* 34 (1978):168–180. The fate of Lewin's psychology after his death represents a very mixed picture and goes beyond the bounds of this volume. For some positive developments, see E. Stivers and S. Wheelan, eds., *The Lewin Legacy: Field Theory in Current Practice* (Berlin: Springer-Verlag, 1986).

Chapter 11. The social construction of psychological knowledge

1 For various elaborations of this general scheme, see R. Whitley, *The Intellectual and Social Organization of the Sciences* (Oxford: Clarendon Press, 1984); also R. Rilling, *Theorie und Soziologie der Wissenschaft* (Frankfurt: Fischer, 1975).

2 This process is extensively described in B. Latour, *Science in Action* (Cambridge, Mass.: Harvard University Press, 1987).

3 An earlier version of this analysis, applied to the origins of modern psychology, is to be found in K. Danziger, "The social origins of modern psychology," in A. R. Buss, ed., *Psychology in Social Context* (New York: Irvington, 1979), pp. 27–45.

4 W. van den Daele, "Die soziale Konstruktion der Wissenschaft-Institutionalisierung und Definition der positiven Wissenschaft in der zweiten Hälfte des 17. Jahrhunderts," in G. Böhme, W. van den Daele, W. Krohn, *Experimentelle Phi-*

losophie: Ursprünge autonomer Wissenschaftsentwicklung (Frankfurt: Suhrkamp, 1977), pp. 129–184.

5 For specific examples from fields other than psychology see Latour, *Science in Action*.

6 The category of “folk science,” introduced by J. R. Ravetz, fits this case exactly: “In each case, the attempt is to reproduce what is believed to be the crucial feature of an established science, where this is learned more from the philosophers of science than from the successful practitioners themselves.” See J. R. Ravetz, *Scientific Knowledge and Its Social Problems* (New York: Oxford University Press, 1971), p. 368. Folk science is marked by the “pretence of maturity,” and extreme responsiveness to external pressures. It should be added that “folk science” is an analytic category, and there is no implication that all branches of a heterogeneous field like psychology fit this category.

7 D. S. Napoli, *The Architects of Adjustment: The History of the Psychological Profession in the United States* (Port Washington, N.Y.: Kennikat Press, 1981).

8 For some key examples see F. Samelson, “Putting psychology on the map: Ideology and intelligence testing,” in A. R. Buss, ed., *Psychology in Social Context* (New York: Irvington, 1980), pp. 103–168; F. Samelson, “Organizing for the kingdom of behavior: Academic battles and organizational policies in the twenties,” *Journal of the History of the Behavioral Sciences* 21 (1985):33–47; J. Morawski, “Organizing knowledge and behavior at Yale’s Institute of Human Relations,” *Isis* 77 (1986):219–242.

9 Several pertinent examples are summarized in S. Shapin, “History of science and its sociological reconstructions,” *History of Science* 20 (1982):157–211. A particularly relevant example is discussed by D. Mackenzie and B. Barnes, “Scientific judgement: The Biometry–Mendelism controversy,” in B. Barnes and S. Shapin, eds., *Natural Order: Historical Studies of Scientific Culture* (Beverly Hills, Calif.: Sage, 1979), pp. 191–208.

10 In other words, there is something artificial about the distinction between the constitution of a cognitive field and the constitution of the group identified with this field. See, e.g., M. Callan, “Struggles and negotiations to define what is problematic and what is not: The sociologic translation,” in K. D. Knorr, R. Krohn, and R. Whitley, eds., *The Social Process of Scientific Investigation* (Dordrecht: Reidel, 1980), pp. 197–219.

11 The inseparable link of ethical and methodological issues has emerged in contemporary discussions of research ethics – e.g., J. G. Adair, T. W. Dushenko, and R. C. L. Lindsay, “Ethical regulations and their impact on research practice,” *American Psychologist* 40 (1985):59–72; D. Baumrind, “Research using intentional deception: Ethical issues revisited,” *American Psychologist* 40 (1985):165–174; H. Schuler, *Ethical Problems in Psychological Research* (New York: Academic Press, 1982); A. Métraux, “Die Verschränkung methodologischer und ethischer Handlungsregeln,” in L. Kruse and M. Kumpf, eds., *Psychologische Grundlagenforschung: Ethik und Recht* (Bern: Huber, 1981), pp. 139–162; B. Harris, “Key words: A history of debriefing in social psychology,” in J. G. Morawski, ed., *The Rise of Experimentation in American Psychology* (New Haven: Yale University Press, 1988), pp. 188–212.

12 J. G. Morawski, “Impossible experiments and practical constructions: The social bases of psychologists’ work,” in Morawski, *The Rise of Experimentation in American Psychology*, pp. 72–93. J. G. Adair and B. Spinner, “Subjects’ access to cognitive processes: Demand characteristics and verbal report,” *Journal for the*

Theory of Social Behaviour 11 (1981):31–52; J. G. Adair, “Phenomenology and experimental research,” a paper presented at the annual meeting of the Canadian Psychological Association, Vancouver, 1987.

13 S. Shapin and S. Schaffer, *Leviathan and the Air-Pump: Hobbes, Boyle and the Experimental Life* (Princeton: Princeton University Press, 1985). On the link between the modern concept of evidence and the social character of witnesses, see also B. J. Shapiro, *Probability and Certainty in Seventeenth-Century England* (Princeton: Princeton University Press, 1983), chap. 5.

14 See E. Freidson, *Professional Powers: A Study of the Institutionalization of Formal Knowledge* (Chicago: University of Chicago Press, 1986); and T. L. Haskell, ed., *The Authority of Experts* (Bloomington: Indiana University Press, 1984).

15 Many facets of this development have been critically described in the writings of Sigmund Koch. See “The nature and limits of psychological knowledge: Lessons of a century qua Science,” in S. Koch and D. E. Leary, eds., *A Century of Psychology as Science* (New York: McGraw-Hill, 1985), pp. 75–97, and other papers by this author listed in the above, pp. 96–97.

16 N. Rose, *The Psychological Complex: Psychology, Politics and Society in England 1869–1939* (London: Routledge and Kegan Paul, 1985).

17 A salient example is provided by historical changes in the vogue of hereditarianism and environmentalism among American psychologists. See H. Cravens, *The Triumph of Evolution: American Scientists and the Heredity–Environment Controversy 1900–1941* (Philadelphia: University of Pennsylvania Press, 1978). It should be noted that in psychology swings of the heredity–environment pendulum were made possible by the fact that both positions shared a common ideological basis; see J. Harwood, “Heredity, environment, and the legitimization of social policy,” in Barnes and Shapin, *Natural Order*, pp. 231–248. A related change of focus occurred in social psychology; see F. Samelson, “From ‘race psychology’ to ‘studies in prejudice’: Some observations on the thematic reversal in social psychology,” *Journal of the History of the Behavioral Sciences* 14 (1978):265–278.

18 There are many accounts of the cultural and historical significance of this myth. An analysis that is of particular interest in the present context is to be found in the first three chapters of R. M. Unger, *Knowledge and Politics* (New York: Free Press, 1975). For a discussion of the psychological aspects, see C. Venn, “The subject of psychology,” in J. Henriques, W. Hollway, C. Urwin, C. Venn, V. Walkerdine, *Changing the Subject* (London: Methuen, 1984), pp. 119–152.

19 For examples, see E. E. Sampson, “Psychology and the American ideal,” *Journal of Personality and Social Psychology* 35 (1977):767–782; and further analysis in E. E. Sampson, “Deconstructing psychology’s subject,” *The Journal of Mind and Behavior* 4 (1983):135–164.

20 R. Harré, *Personal Being* (Cambridge, Mass.: Harvard University Press, 1984); A. Pepitone, “Lessons from the history of social psychology,” *American Psychologist* 36 (1981):972–985.

21 K. J. Gergen, *Toward Transformation in Social Knowledge* (New York: Springer, 1982); R. Harré and P. F. Secord, *The Explanation of Social Behaviour* (Oxford: Blackwell, 1972); K. Holzkamp, “Selbsterfahrung und wissenschaftliche Objektivität: Unaufhebbarer Widerspruch?” in K.-H. Brown and K. Holzkamp, eds., *Subjektivität als Problem psychologischer Methodik* (Frankfurt and New York: Campus, 1985), pp. 17–37.

22 J. D. Greenwood, “On the relation between laboratory experiments and social behavior: Causal explanation and generalization,” *Journal for the Theory of Social*

Behaviour 12 (1982):225–250. Analogous problems arise in the currently popular psychology of cognition; see J. Lave, *Cognition in Practice* (New York: Cambridge University Press, 1988).

23 For example, intelligence tests are best at predicting performance in scholastic settings, group performance tasks do relatively well at predicting in group problem-solving situations; in general, what psychological tests do best is predict performance on other psychological tests. On the problems of psychological prediction, see L. J. Cronbach, "Beyond the two disciplines of scientific psychology," *American Psychologist* 30 (1975):116–127. On the question of the relationship between the investigative situation and other social situations, see K. Holzkamp, "Zum Problem der Relevanz psychologischer Forschung für die Praxis," *Psychologische Rundschau* 21 (1970):1–22.

24 The classical statement of the distinction within the context of American psychology is G. W. Allport, *Personality: A Psychological Interpretation* (New York: Holt, Rinehart & Winston, 1937). It should be noted that, although the terms were borrowed from the neo-Kantian philosopher, Windelband, their reference had undergone a characteristic change, the original historical context having been replaced by an ahistorical one.

25 See B. Latour, "Give me a laboratory and I will raise the world," in K. Knorr-Cetina and M. Mulkay, eds., *Science Observed: Perspectives on the Social Study of Science* (Beverly Hills, Calif.: Sage, 1983), pp. 141–170.

26 B. F. Skinner, *Walden Two* (New York: Macmillan, 1948). The distinctive nature of Skinner's utopia is that it is *imposed*. In the real world, psychology may participate in the reconstruction of social settings through its subtle influence on the social consciousness of participants. See A. MacIntyre, "How psychology makes itself true or false," in Koch and Leary, *A Century of Psychology*, pp. 897–903.

27 This delay is all the more remarkable because the point had been made in a prominent journal in 1933. See S. Rosenzweig, "The experimental situation as a psychological problem," *Psychological Review* 40 (1933):337–354. Subsequently, about three decades elapsed before anyone else was ready to take up the issue in print.

28 For an in vivo study of this ideology, see J. Potter and M. Mulkay, "Making theory useful: Utility accounting in social psychologists' discourse," *Fundamenta Scientiae* 3 (1982):259–278.

29 M. Foucault, *Discipline and Punish: The Birth of the Prison* (New York: Pantheon, 1978), pt. 3 (from the French original, 1975). For psychology's contributions as "discipline," see N. Rose, "Calculable minds and manageable individuals," *History of the Human Sciences* 1 (1988):179–200.

30 For a detailed demonstration of the dependence of one set of investigative practices on historically contingent industrial institutions, see B. Schwartz, R. Schuldenfrei, and H. Lacey, "Operant psychology as factory psychology," *Behaviorism* 6 (1978):229–254; also S. Kvale, "The technological paradigm of psychological research," *Journal of Phenomenological Psychology* 3 (1973):143–159.

31 "External validity" is defined in D. T. Campbell and J. C. Stanley, *Experimental and Quasi-experimental Designs for Research* (Chicago: Rand McNally, 1963).

32 During the latter part of the period that is the focus of this study, these ideological functions became increasingly concentrated in the systems of neobehaviorism. See L. D. Smith, *Behaviorism and Logical Positivism* (Stanford: Stanford University Press, 1986); J. A. Mills, "Hull's theory of learning: II. A criticism of the theory and its relationship to the history of psychological thought," *Canadian*

Psychological Review 19 (1978):116–127; J. A. Mills, “Some observations on Skinner’s moral theory,” *Journal for the Theory of Social Behaviour* 12 (1982):140–160; Morawski, “Organizing knowledge.”

33 J. Shotter, *Images of Man in Psychological Research* (London: Methuen, 1975).

34 R. Bhaskar, *A Realist Theory of Science* (Atlantic Highlands, N.J.: Humanities Press, 1978). The implications of this position for the human sciences are discussed in R. Bhaskar, *The Possibility of Naturalism* (Atlantic Highlands, N.J.: Humanities Press, 1979); P. T. Manicas, “The human sciences: A radical separation of psychology and the social sciences,” in P. F. Secord, ed., *Explaining Human Behavior: Consciousness, Human Action and Social Structure* (Beverly Hills, Calif.: Sage, 1982), pp. 155–173; P. T. Manicas and P. F. Secord, “Implications for psychology of the new philosophy of science,” *American Psychologist* 38 (1983):399–413.

35 The domain of the real in this sense should not be thought of as a domain of entities but as a domain of causal powers. See I. Hacking, *Representing and Intervening* (Cambridge: Cambridge University Press, 1983), chaps. 2 and 12; also R. Harré and E. H. Madden, *Causal Powers* (Totowa, N.J.: Rowman and Littlefield, 1975).

36 The specifics of Wundt’s notion of psychic causality cannot be explored here. See his statements in his *Grundzüge der physiologischen Psychologie*, 5th ed., vol. 3 (Leipzig: Engelmann, 1903), and his *Logik*, 3d ed., vol. 3 (Stuttgart: Enke, 1908), pp. 243–295. For an earlier statement in English, see his *Outlines of Psychology* (1897), sec. 5, republished in R. W. Rieber, ed., *Wilhelm Wundt and the Making of a Scientific Psychology* (New York: Plenum, 1980). The significance of the concept of psychic causality for Wundt’s experimental practice is discussed in K. Danziger, “Wundt’s psychological experiment in the light of his philosophy of science,” *Psychological Research* 42 (1980):109–122. See also T. Mischel, “Wundt and the conceptual foundations of psychology,” *Philosophy and Phenomenological Research* 31 (1970):1–26.

37 A particularly far-reaching account of Gestalt principles as a form of psychological causality is presented in W. Köhler, *The Place of Value in a World of Facts* (New York: Liveright, 1938).

38 S. Toulmin and D. E. Leary, “The cult of empiricism in psychology, and beyond,” in Koch and Leary, *Psychology as Science*, pp. 594–617; K. Danziger, “The positivist repudiation of Wundt,” *Journal of the History of the Behavioral Sciences* 15 (1979):205–230; B. D. Mackenzie, *Behaviourism and the Limits of Scientific Method* (Atlantic Highlands, N.J.: Humanities Press, 1977).

39 Scientific activity commonly involves a two-sided procedure: In one phase the empirical domain is accepted and the theoretical domain is put in doubt, but in the other phase the theoretical domain is accepted and the empirical domain is put in doubt. See R. G. Francis, *The Rhetoric of Science* (St. Paul: University of Minnesota Press, 1961). If all the emphasis is placed on either one of these phases, one gets either a jumble of disjointed facts or a list of speculative systems but not a developing science.

40 As Boyd points out, in science “theoretical considerations are evidential.” See R. N. Boyd, “The current status of scientific realism,” in J. Leplin, ed., *Scientific Realism* (Berkeley: University of California Press, 1984), p. 61.

41 Because alternative conceptions of psychological reality are involved in different types of investigative practice, and therefore lead to different types of data, the confrontation of investigative practice with a divergent theoretical order can sometimes take the form of a confrontation of empirical data. During the period covered

by the present volume, an important example of this type of confrontation was provided by the comparison of Gestalt psychological experiments on problem solving in primates (W. Köhler, *The Mentality of Apes* (New York: Harcourt, Brace, 1925), with American experiments on cats in puzzle boxes and rats in mazes. Here the empirical confrontation served as the occasion for the critique of certain investigative practices from the perspective of a divergent conception of psychological reality. However, such confrontations in depth were the exception rather than the rule. Typically, the original confrontation was rendered harmless by its transformation into the issue of continuity versus discontinuity in learning that flourished in the 1930s and beyond (K. W. Spence, "The nature of discrimination learning in animals," *Psychological Review* 43 [1936]:427–449). With this transformation the potentially disturbing features of the original critique were successfully avoided and the dominant style of investigative practice could continue to be taken for granted. The issue became trivialized to a form of question that could be investigated within the framework of traditional practices. This was the usual fate of controversies that threatened to upset the methodological applecart.

42 For America's first Ph.D. in psychology, J. Jastrow, psychology was already "the master science of the human group," and "the psychologist has come naturally and without presumption to assume the role of helmsman" (*Piloting Your Life: The Psychologist as Helmsman* [Garden City, N.Y.: Greenberg, 1930], p. xiii); E. L. Thorndike (in 1907!) declared that psychology "supplies or should supply the fundamental principles upon which sociology, economics, history, anthropology, linguistics and the other sciences dealing with human thought and action should be based" (*The Elements of Psychology* [New York: Seiler, 1907] p. 323); J. McKeen Cattell added that "it is also for psychology to determine what does in fact benefit the human race" ("The usefulness of psychology," *Science* 72 [1930]:284–287). Although such arrogance could not command universal assent, it did reflect the confidence of men who were securely grounded in the popular prejudices of their age.

43 P. T. Manicas, *A History and Philosophy of the Social Sciences* (Oxford: Blackwell, 1987), pp. 261–265.

44 The general position that "the interplay between 'criteria', 'rules', 'methodological strictures', and the 'cognitive content' of science is constant and tight" is developed extensively in D. Shapere, *Reason and the Search for Knowledge* (Dordrecht: Reidel, 1984) (quotation on p. 216); see also the discussion of an earlier statement of this view by T. Nickels, "Heuristics and justification in scientific research," in F. Suppe, ed., *The Structure of Scientific Theories* (Urbana: University of Illinois Press, 1974), pp. 571–589.

45 D. Bloor, *Wittgenstein: A Social Theory of Knowledge* (New York: Columbia University Press, 1983), chaps. 3 and 7.

46 G. Lakoff, *Women, Fire and Dangerous Things: What Categories Reveal about the Mind* (Chicago: University of Chicago Press, 1987).

47 R. W. Miller, *Fact and Method* (Princeton: Princeton University Press, 1987); R. Harré, *Varieties of Realism* (Oxford: Blackwell, 1986); M. G. Raskin and H. J. Bernstein, *New Ways of Knowing: The Sciences, Society, and Reconstructive Knowledge* (Totowa, N.J.: Rowman and Littlefield, 1987).

48 K. Danziger, "Psychological objects, practice, and history," *Annals of Theoretical Psychology* 8 (New York: Plenum, in press.).

49 "It is necessary for us to see that of the two main ways in which we may explain the growth of science, one is rather unimportant and the other is important. The

first explains science by the accumulation of knowledge: it is like a growing library (or a museum). As more and more books accumulate, so more and more knowledge accumulates. The other explains it by criticism: it grows by a method more revolutionary than accumulation – by a method which destroys, changes, and alters, the whole thing including its most important instrument, the language in which our myths and theories are formulated.” See K. R. Popper, *Conjectures and Refutations: The Growth of Scientific Knowledge* (New York: Harper & Row, 1968), p. 129.

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